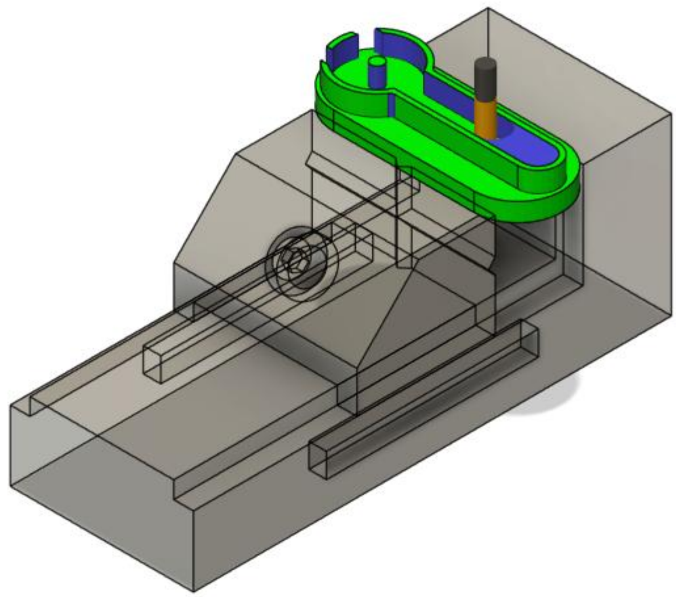
Desktop CNC Milling Machine: Cutting Crank-Slider BASE

I: General Guidelines



GENERAL SAFETY AND OPERATIONAL INSTRUCTIONS – READ BEFORE USE:

- Users must demonstrate proper personal protective equipment (PPE) use before performing work on this machine.
- Users must wear appropriate PPE (safety glasses, ear protection and mask when necessary) when operating this machine.
- NEVER operate the machine tool without all safety mechanisms (enclosure, door switch, emergency stop, limit switches) connected, complete, and fully operational.
- **NEVER BYPASS ANY SAFETY EQUIPMENT.**
- NEVER place hands, head, or other body parts inside the machine tool enclosure while spindle is powered and door switch / emergency stop are disengaged.
- NEVER handle sharp tools in the spindle without the emergency stop engaged.
- During ANY milling operation, if the cutting tool shows excessive wear or excessive chip build up, or if the spindle's RPM changes noticeably (this is often audible), quickly feed hold or emergency stop the machine (depending on severity) to inspect and determine if it is safe to continue.
- Always allow the spindle to reach its target speed before entering a cut.
- Always double-check setup steps and part programming to minimize machine accidents.
- Always watch and listen to EVERY milling operation closely. Keep a hand resting over the emergency stop button: quick reactions are necessary in mitigating damage to the machine and tools.

II: Operator Checklist

Operator name: ______________________________________

Workpiece:____

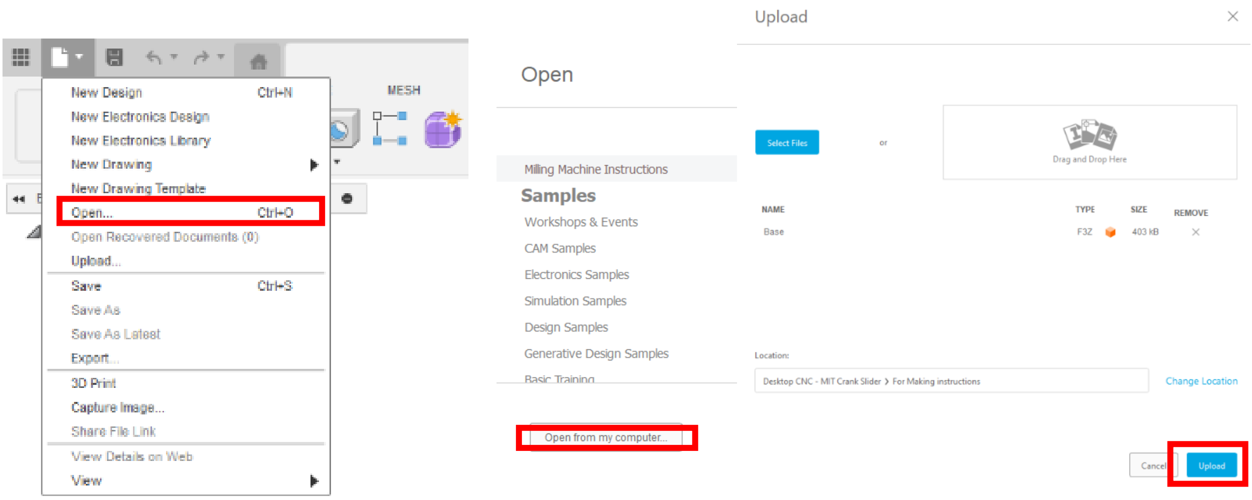
Date:____

Before running the program		
OP10	OP20	
☐	☐	Program has been simulated in CAM
☐	☐	All tool offsets are set per the setup sheet
☐	☐	Workpiece is clamped in the proper position
☐	☐	Correct program is selected on the machine
		Work offsets are set for:
☐	☐	X axis
☐	☐	Y axis
☐	☐	Z axis

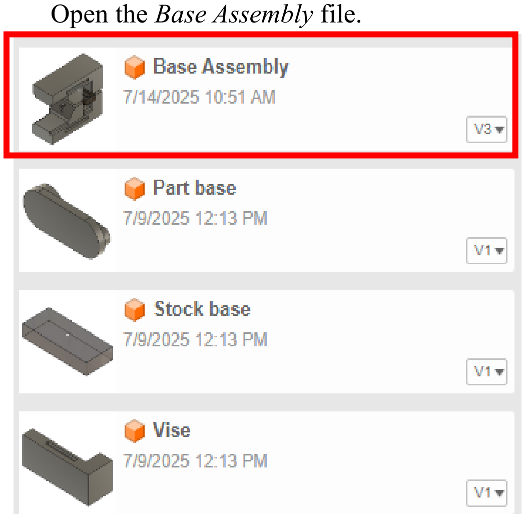
For each tool while running the program		
OP10	OP20	
☐	☐	Check position as the tool approaches the part

III. Part Programming (Autodesk Fusion) OP10

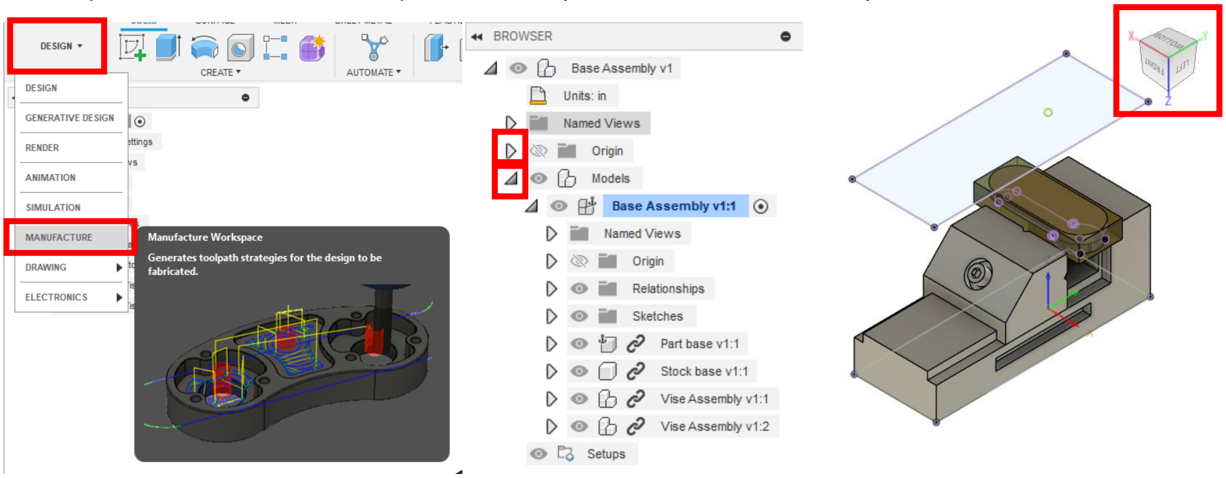
1. Start Fusion and open the provided file. Select the open option from the drop-down menu in the top left corner of Fusion. Select *Open from my computer…* in the following prompt. Find *Base Assembly.f3z* and select *upload*.



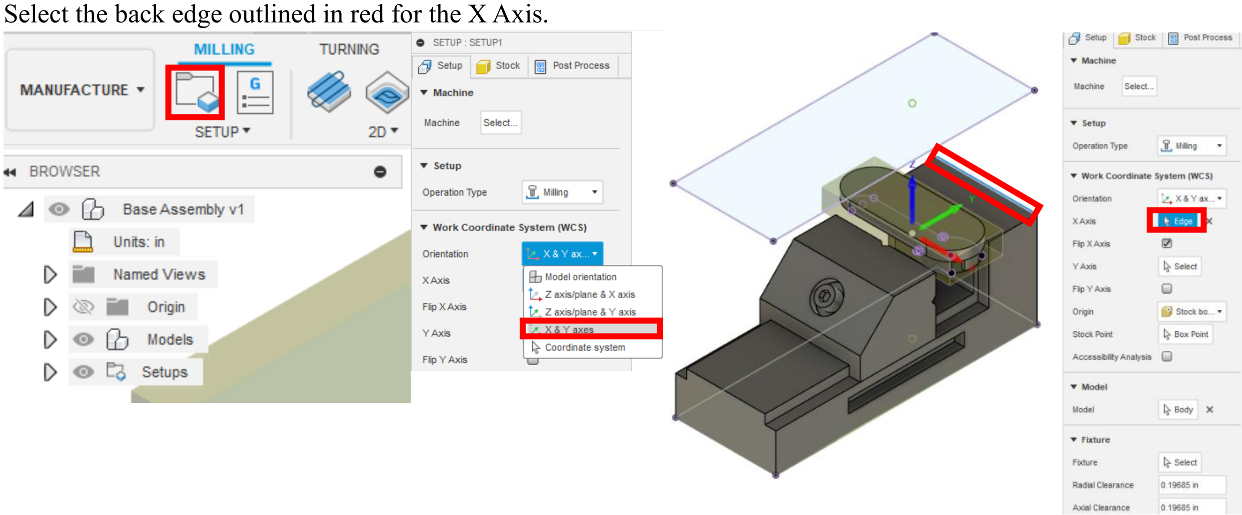
2. After uploading the file, three parts and an assembly will be in the folder on the left-hand side of the program. Open the *Base Assembly* file.



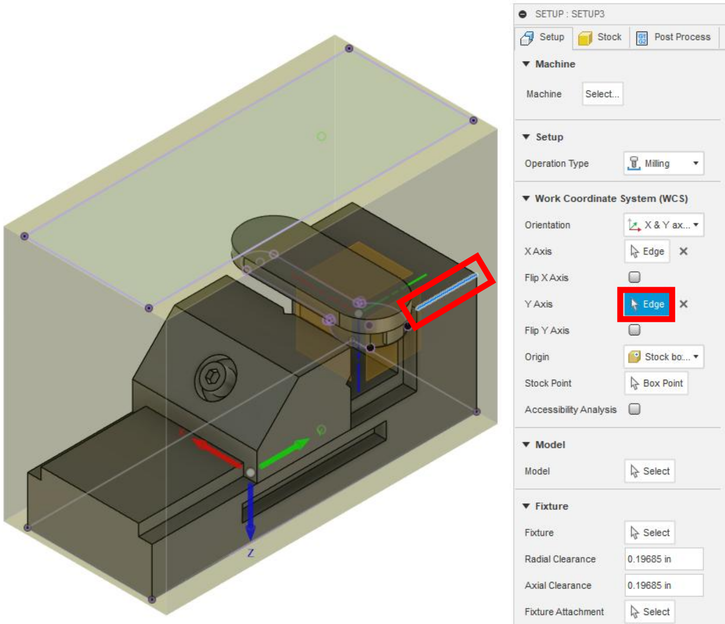
3. From the drop-down menu, select the Manufacture tab. In browser, drop down *Models* then drop down *Base Assembly v1:1* and hide *Vice Assembly v1:2*. Ensure your view cube and assembly are oriented as shown.



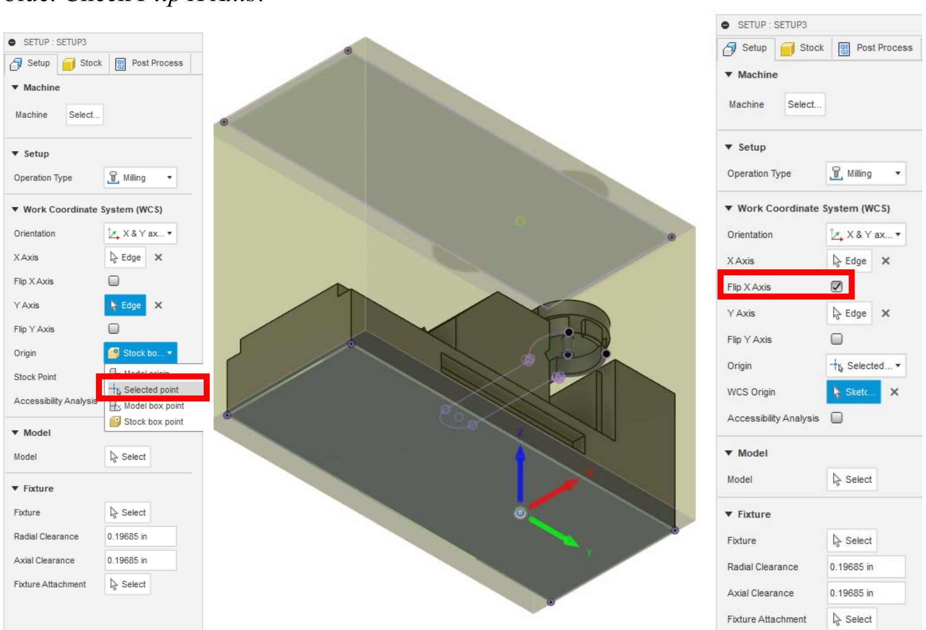
4. To begin toolpath generation, create a new *Setup*. From the Orientation drop-down menu, select X & Y axes Select the back edge outlined in red for the X Axis.



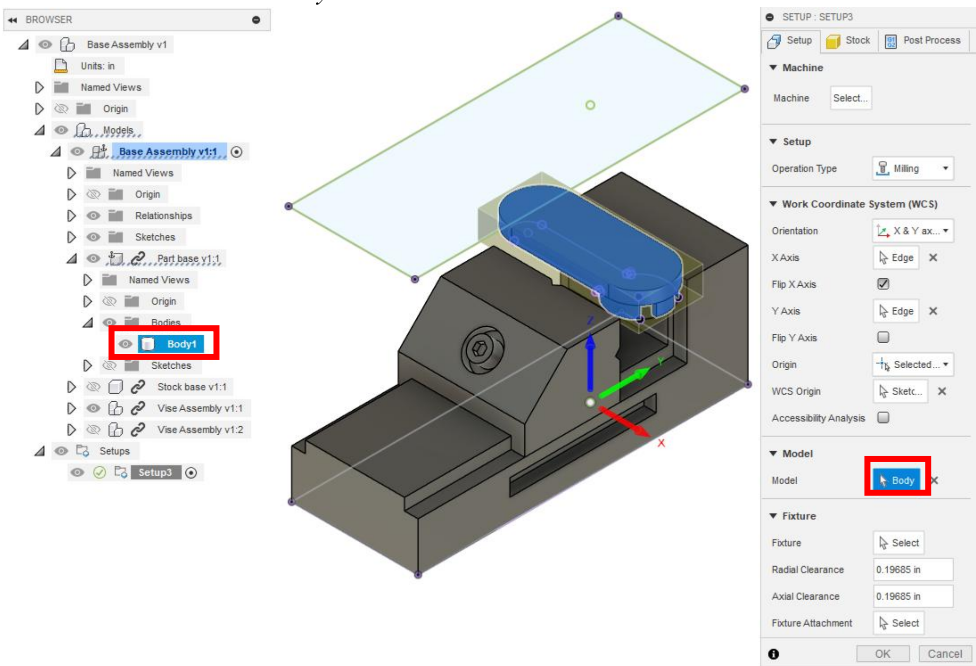
5. After selecting the X axis, select the Y axis as the perpendicular edge outlined in red.



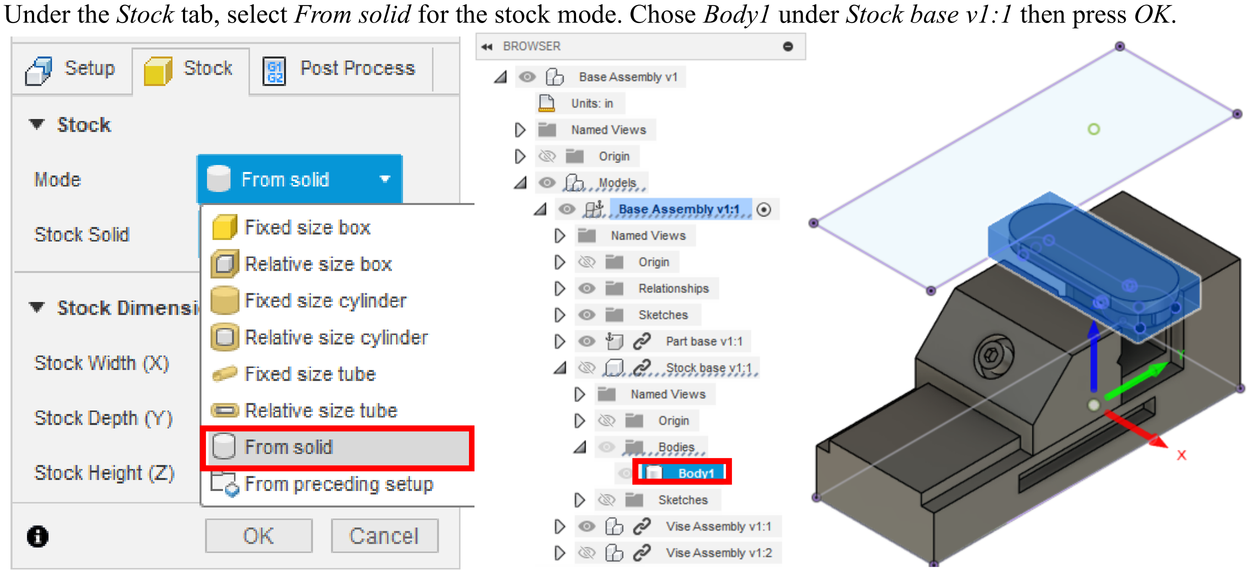
6. Select *Selected point* under the Origin drop-down menu. Select the point on the bottom of the vise outlined in blue. Check *Flip X Axis*.



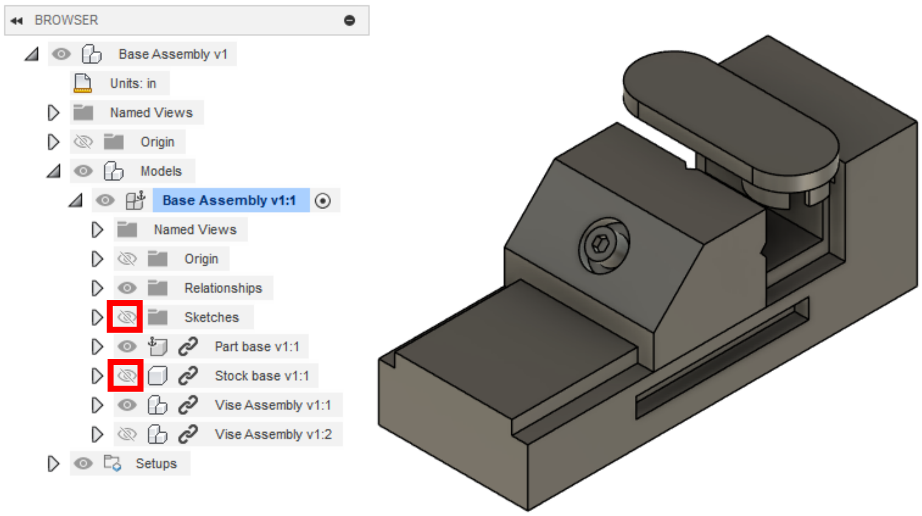
7. Select *Model* and choose *Body1* under *Part base v1:1*.



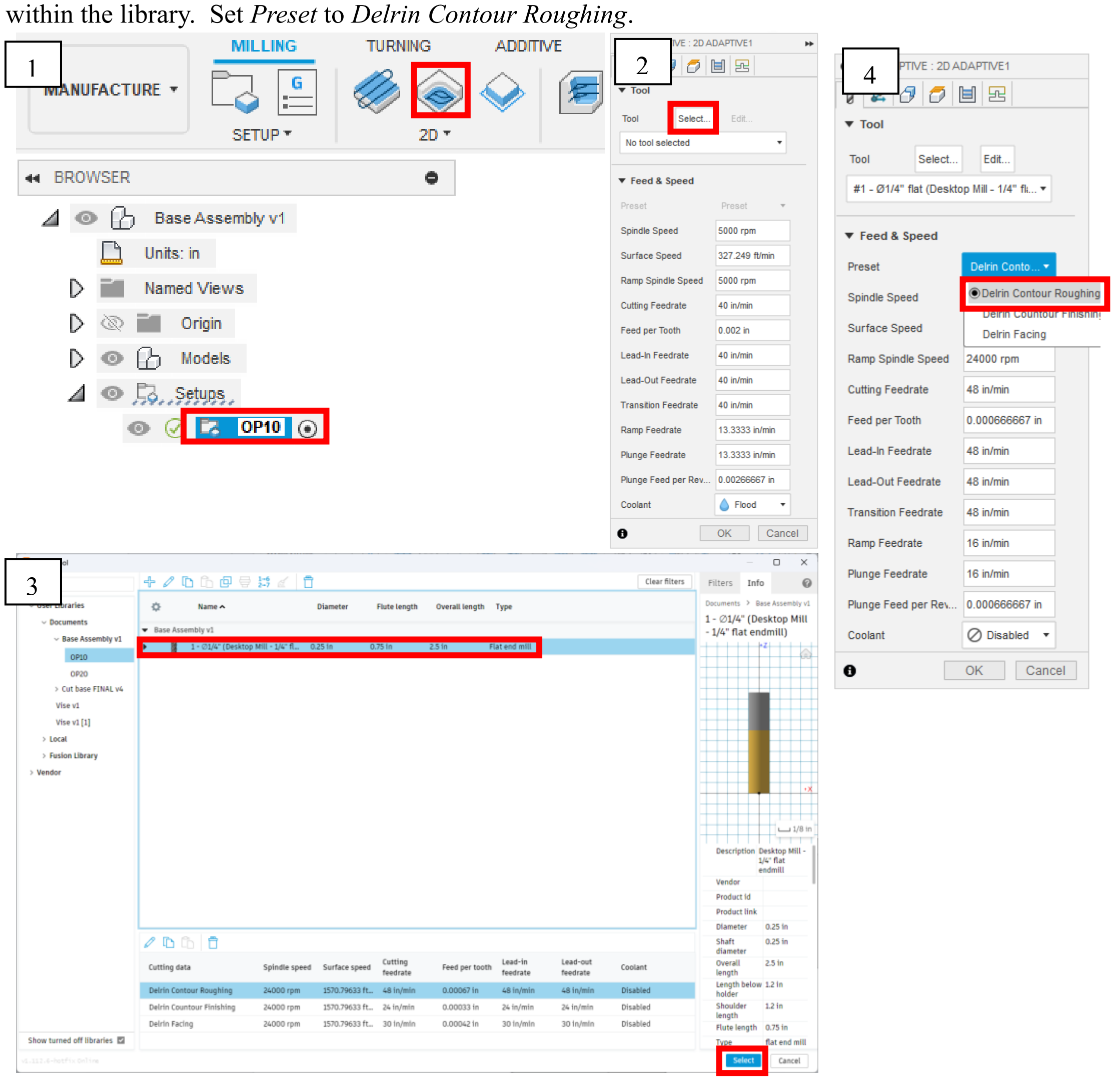
8. Under the *Stock* tab, select *From solid* for the stock mode. Chose *Body1* under *Stock base v1:1* then press *OK*.



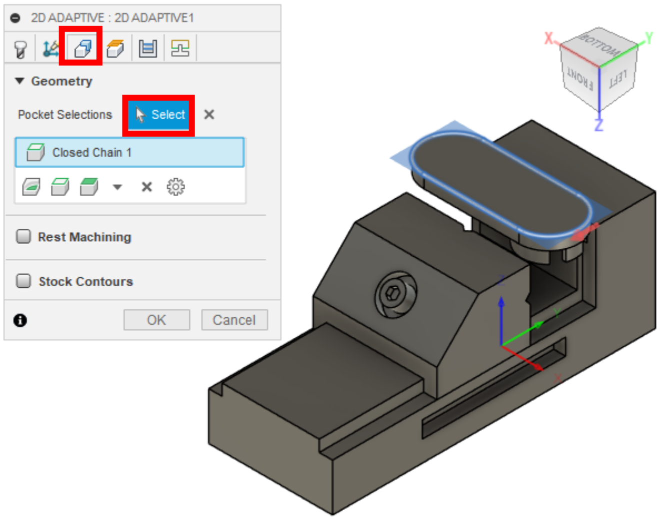
9. Hide the *Sketches* under *Base Assembly v1:1*. Hide the component *Stock base v1:1*. To hide components and sketches select the eye outlined in red.



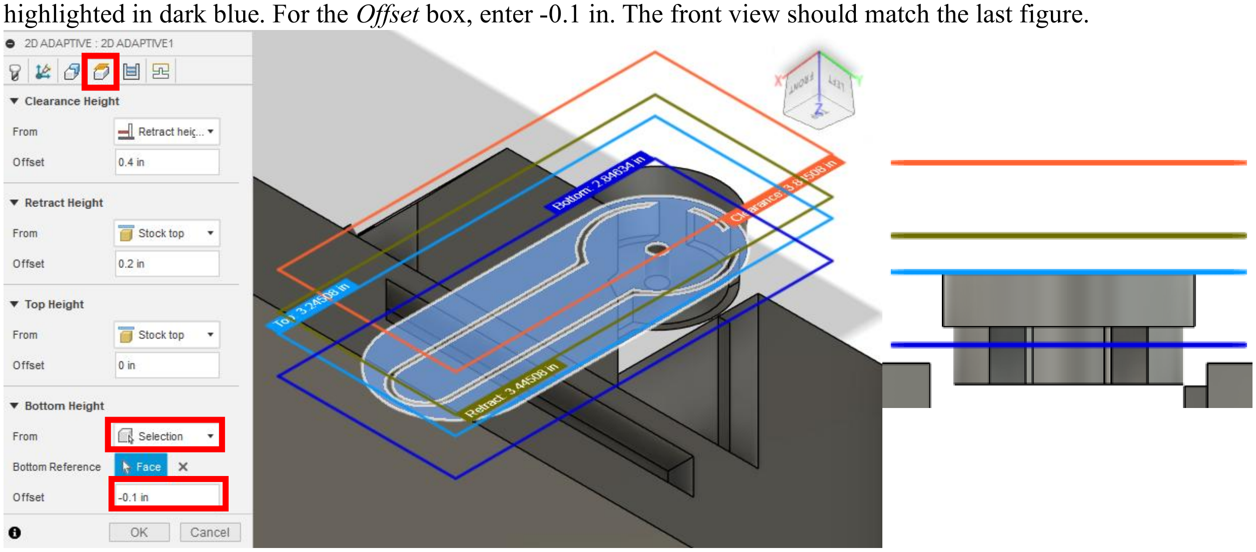
10. Slowly double click *Setup1* and rename to *OP10*. Next select *2D Adaptive Clearing* and select *1/4" flat endmill* within the library. Set *Preset* to *Delrin Contour Roughing*.



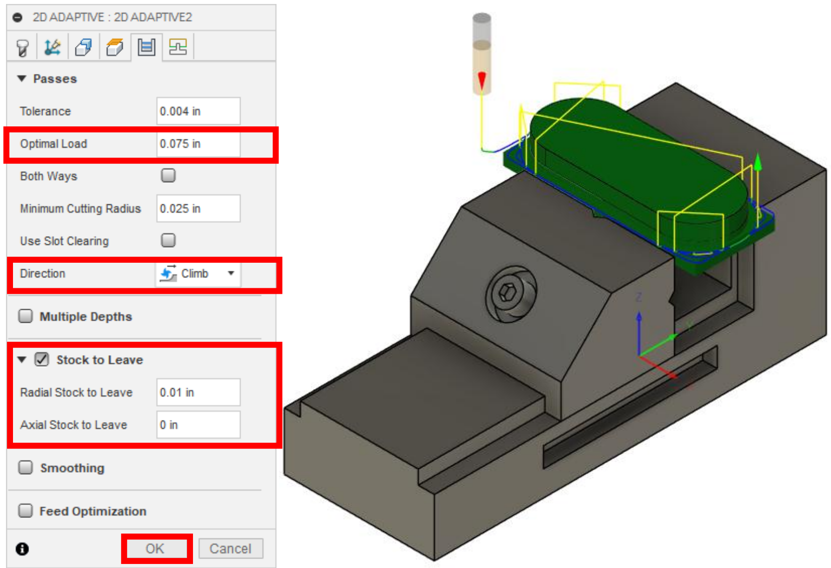
11. Select the *Geometry* tab and for pocket selections select the outer edge of the part outlined in blue. Ensure that the red arrow is outside of the part and in the correct direction as shown below.



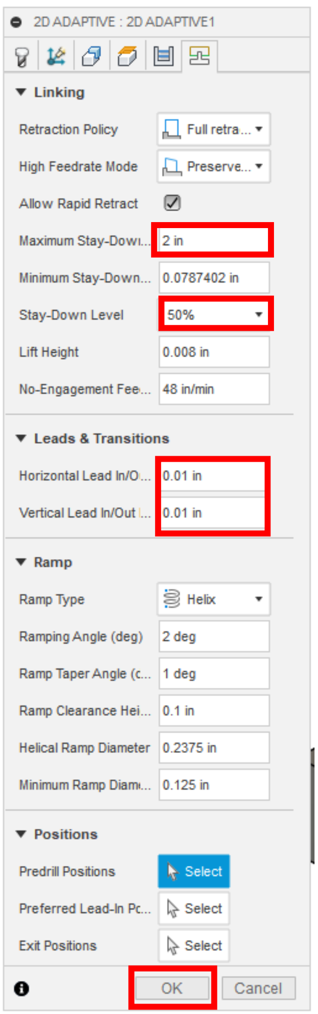
12. Select the *Heights* tab and for the *Bottom Height* Drop down select *Selection*. Select the bottom surface highlighted in dark blue. For the *Offset* box, enter -0.1 in. The front view should match the last figure.



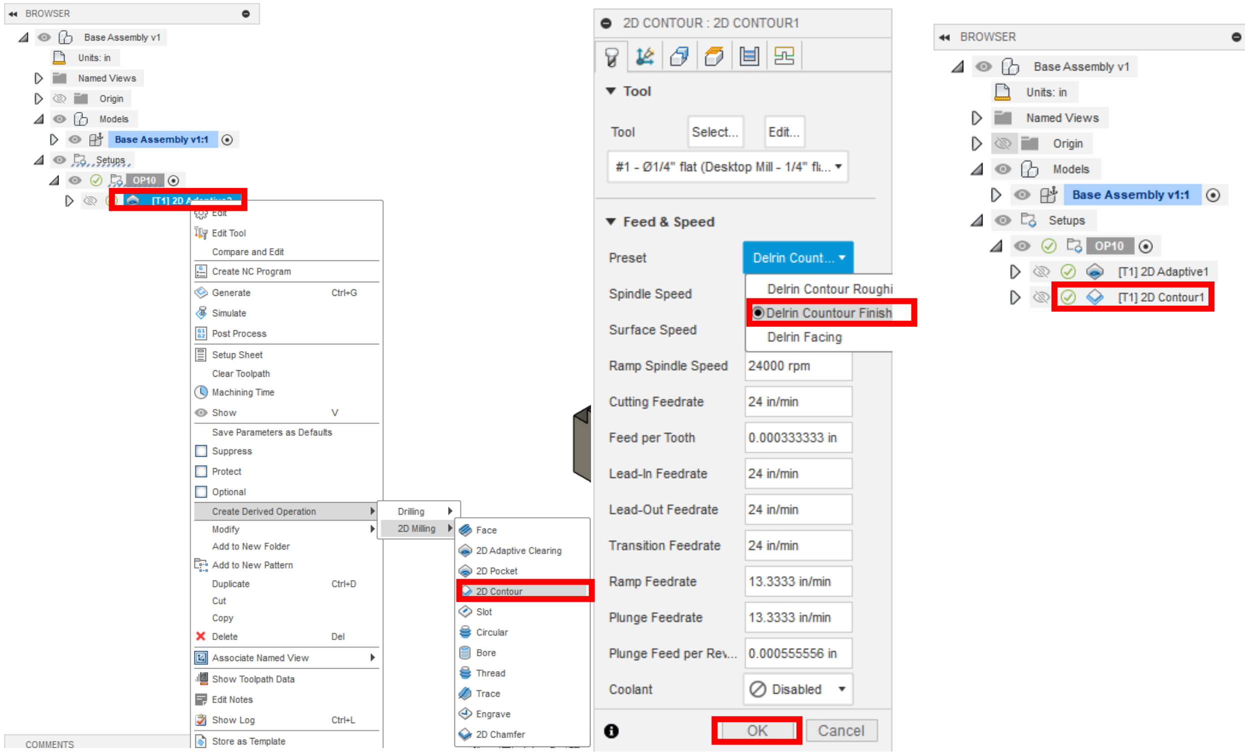
13. Under the *Passes* tab, enter 0.075 in for the *Optimal Load.* Ensure the *Direction* is set to *Climb* and *Radial Stock to Leave* is set to 0.01 in. *Axial Stock to Leave* should be set to zero. Select *OK* when finished.



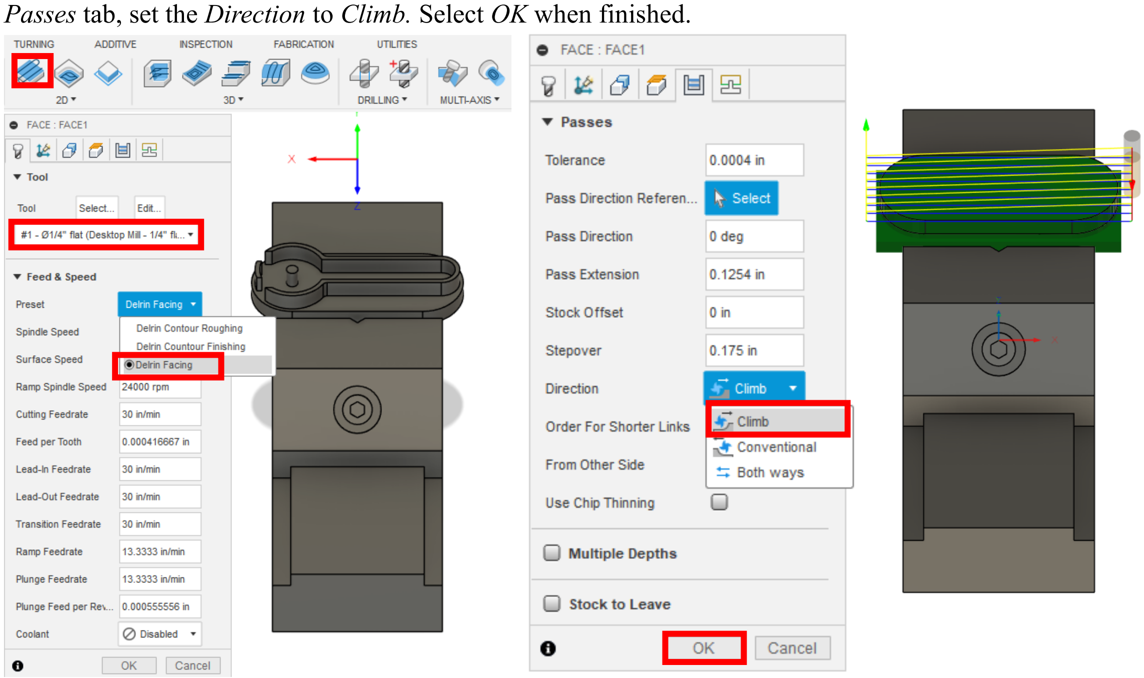
14. Under the *Linking* tab, set*Maxiumum Stay-Down Distance* to 2 in, *Stay-Down Level* to 50%, and *Horizontal and Vertical LeadIn/out Radius to 0.01 in.* Press *OK.*



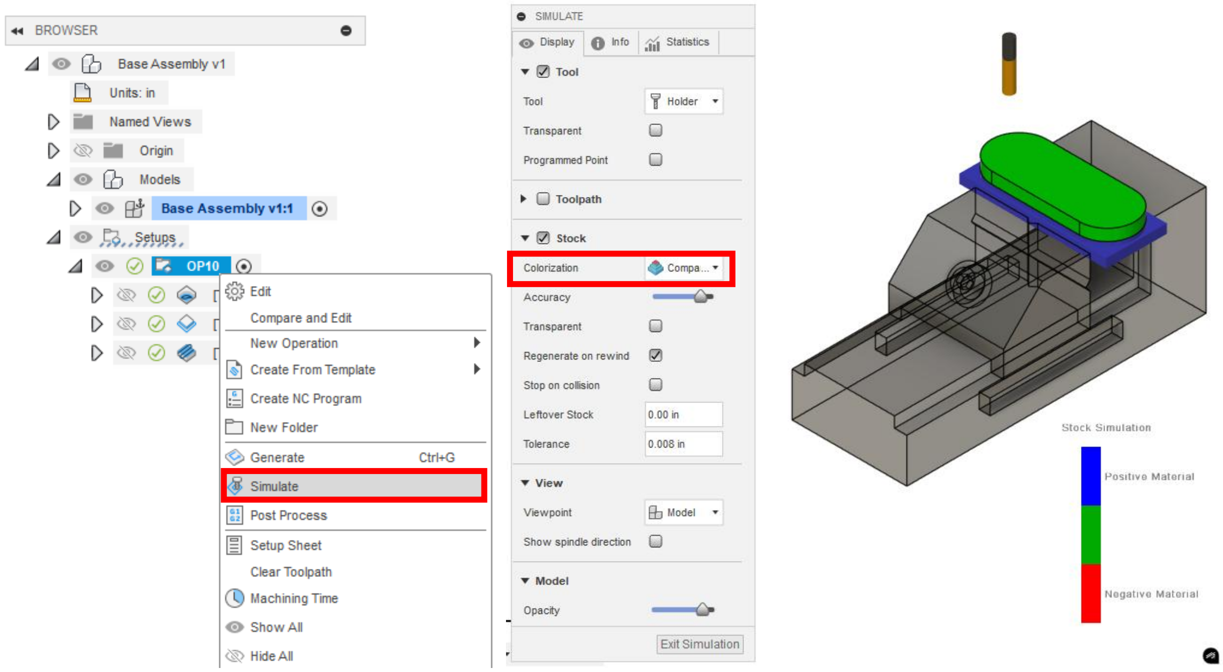
15. Right click the *2D Adaptive Clearing* tool path that was just created. Select *Create Derived Operation > 2D Milling > 2D Contour.* Select the *Delrin Contour Finishing* preset.



16. Select *Face* and ensure the same tool is selected. For the *Feed & Speed* preset, use *Delrin Facing*. Under the *Passes* tab, set the *Direction* to *Climb.* Select *OK* when finished.

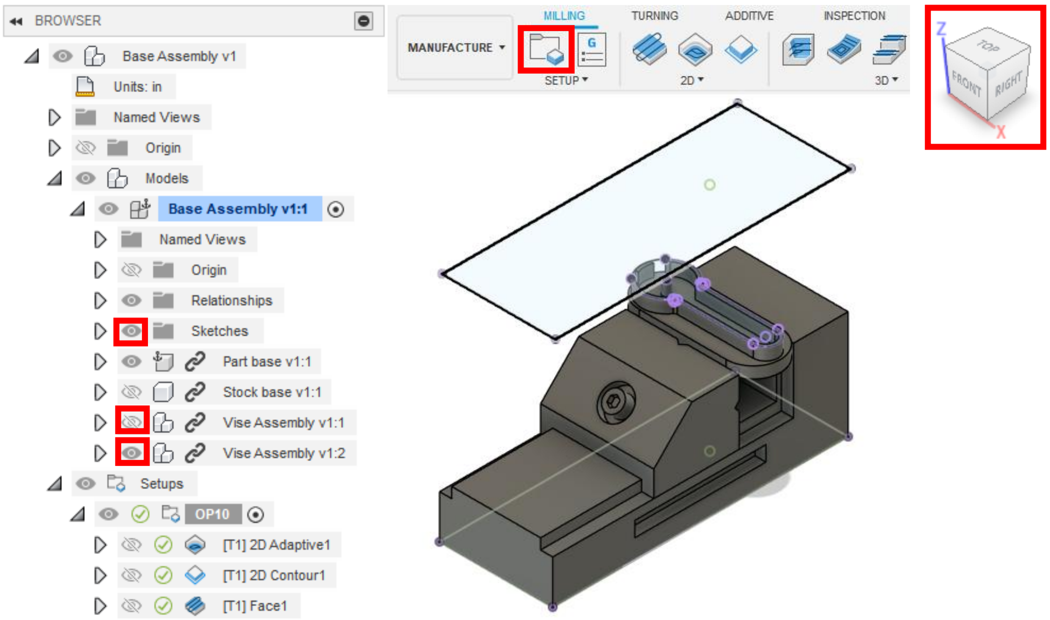


17. *OP10* should be finished at this point. Right click on *OP10* and select *Simulate*. Under *Colorization* choose *Comparison.* Press the play button at the bottom center of the window. Adjust the *Accuracy* slider until simulation runs smoothly. Once the simulation is complete, the part should be green if programmed correctly. <u>Note: If *Accuracy* is set to the lowest value, blue may appear around the part.</u>

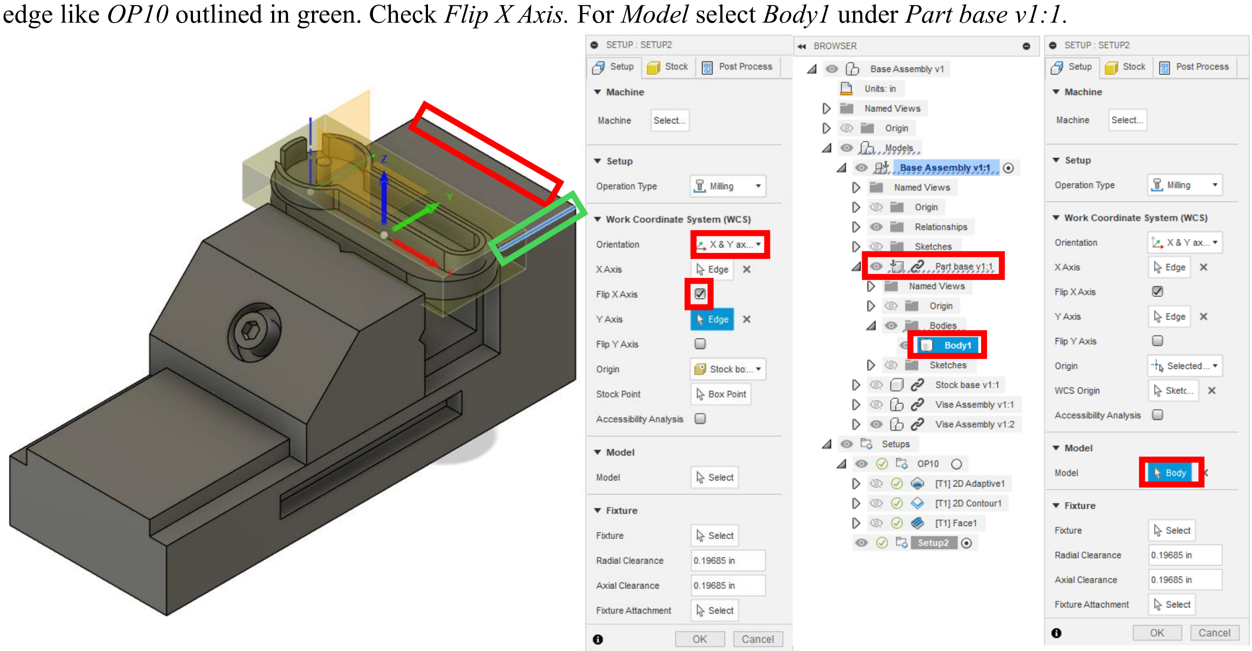


Part Programming (Autodesk Fusion) OP20

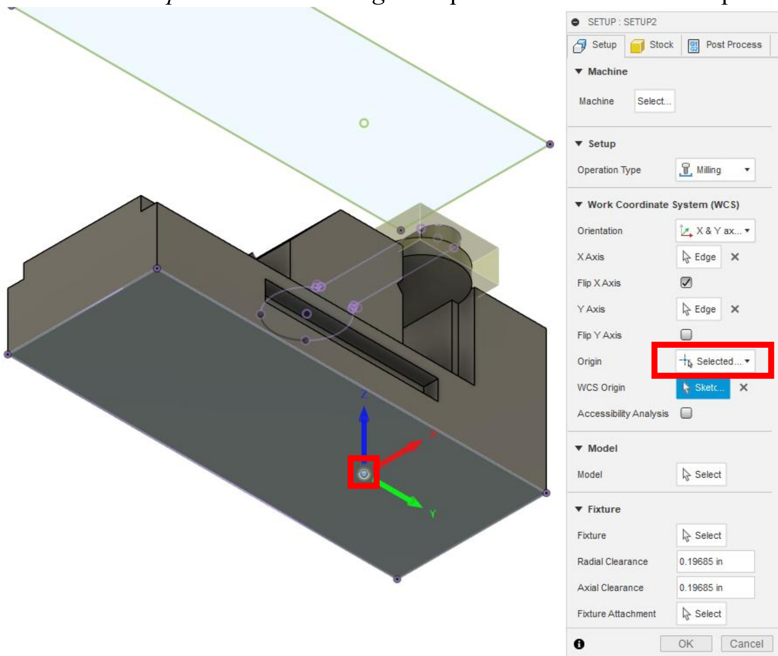
18. Unhide *Sketches* under *Base Assembly v1: 1*. Hide *Vise Assembly v1:1* and unhide *Vise Assembly v1:2* to begin programming OP20. Create a new setup. Ensure that your view block and assembly are oriented in the correct direction.



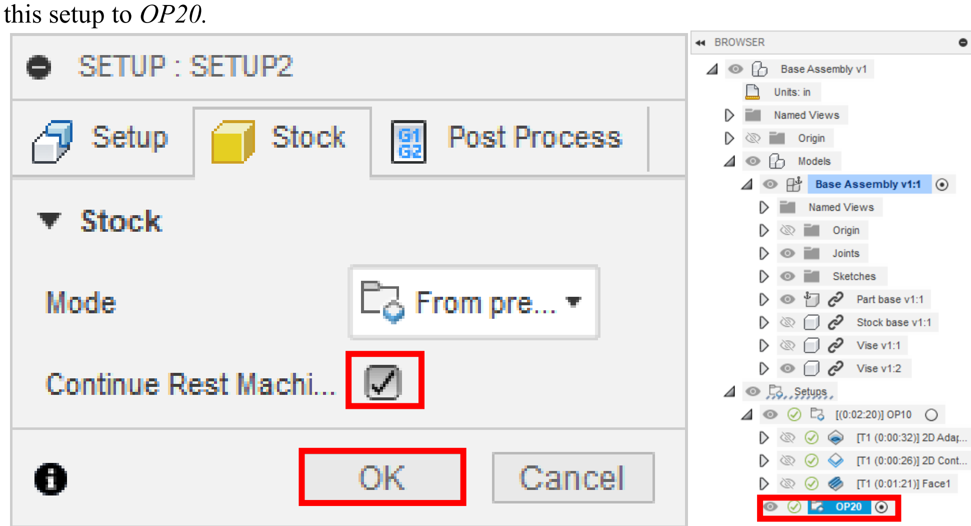
19. For *Orientation* select *X & Y axes* and choose the back edge for X outlined in blue. Select the perpendicular edge like *OP10* outlined in green. Check *Flip X Axis.* For *Model* select *Body1* under *Part base v1:1*.



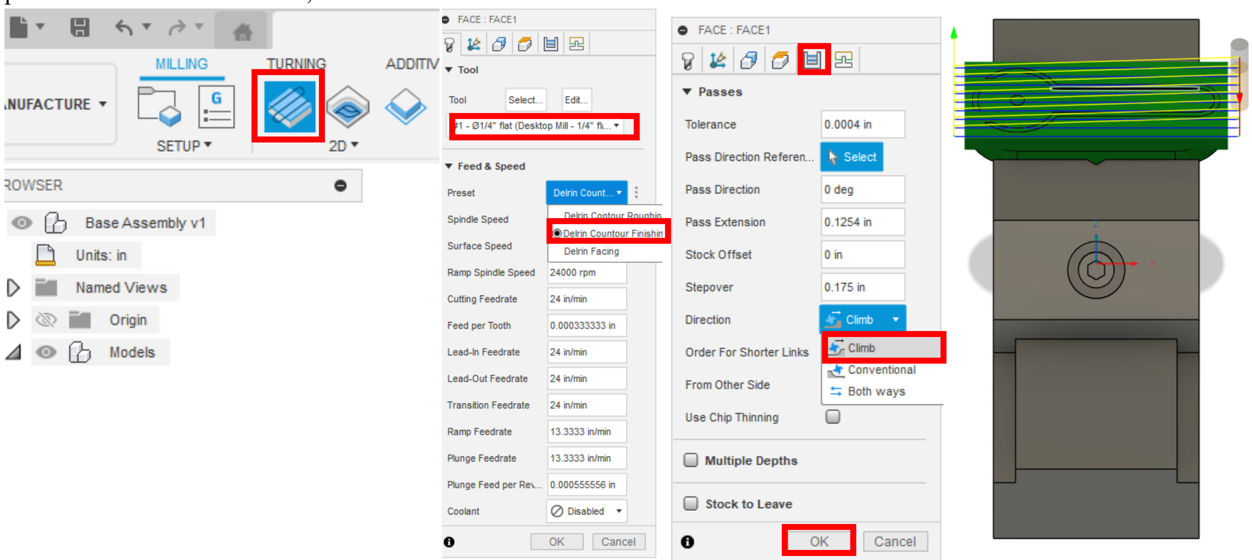
20. Select *Selected point* under the *Origin* drop-down menu. Select the point on the bottom of the vise boxed in red.



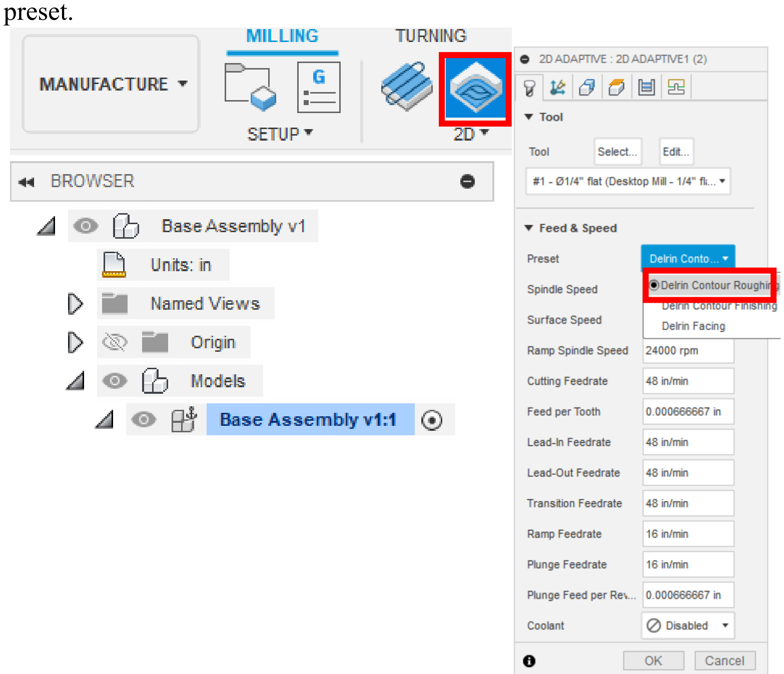
21. Under the *Stock* Tab ensure the *Mode* is set to *From preceding setup.* Check *Continue Rest Machining.* Rename this setup to *OP20.*



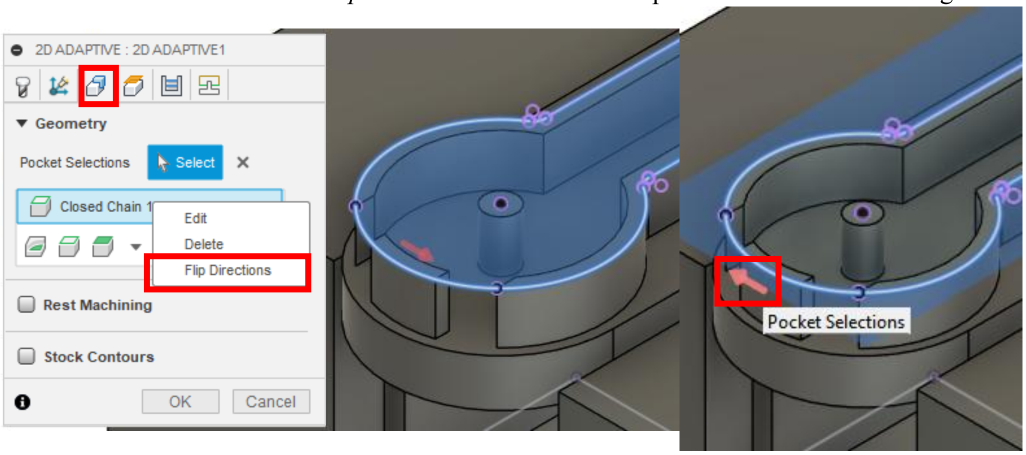
22. Create a *Face* operation. Ensure the same *1/4" flat endmill* is selected. Choose the *Delrin Contour Finishing* preset. Under the *Passes* tab, set the *Direction* to *Climb*.



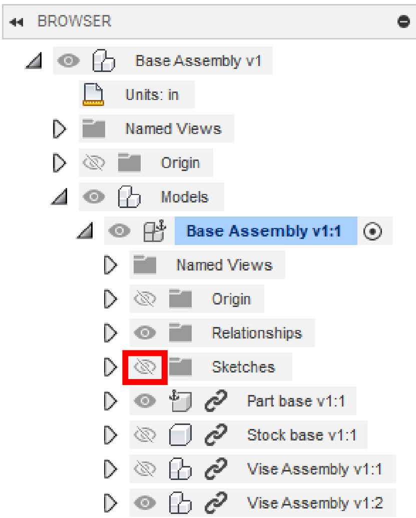
23. Create a *2D Adaptive Clearing* operation. Ensure the same tool is selected. Select the *Delrin Contour Roughing* preset.



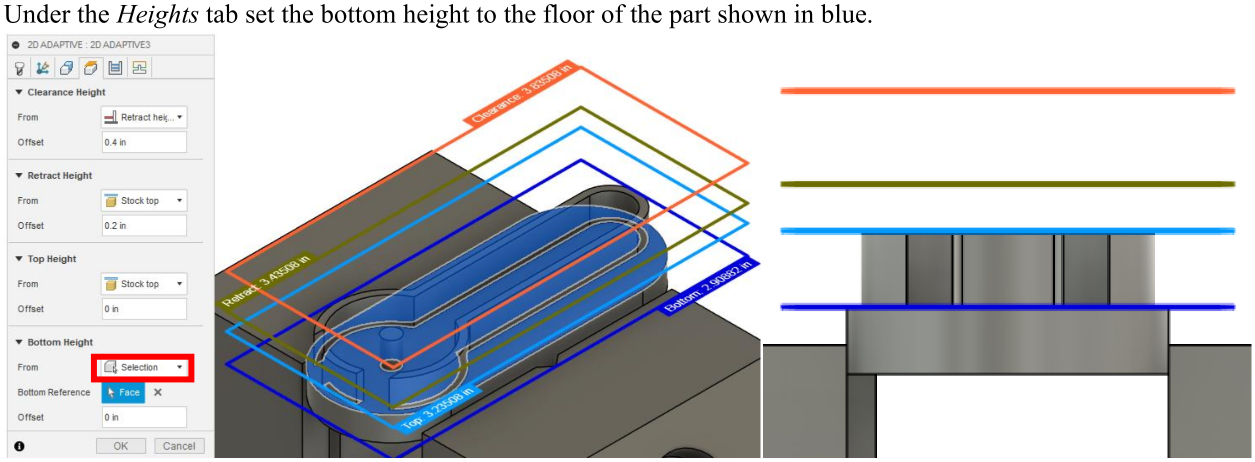
24. Under the *Geometry* tab select the black sketch line around the top edge of the part outlined in blue. Right click on *Closed Chain 1* and select *Flip Directions* so that the tool path follows the outside edge.



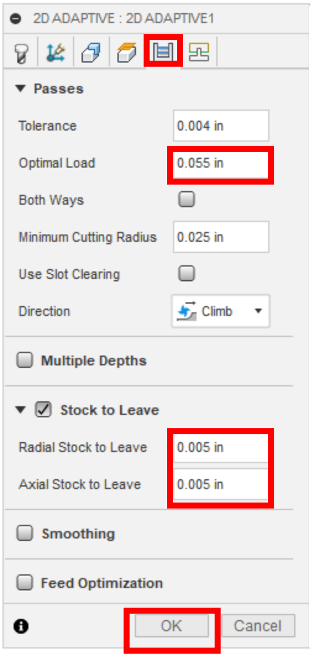
25. Hide the *Sketches* under *Base Assembly v1:1*. Note: If the sketch isn't hidden at this step the selection of the height in the next step will not work properly

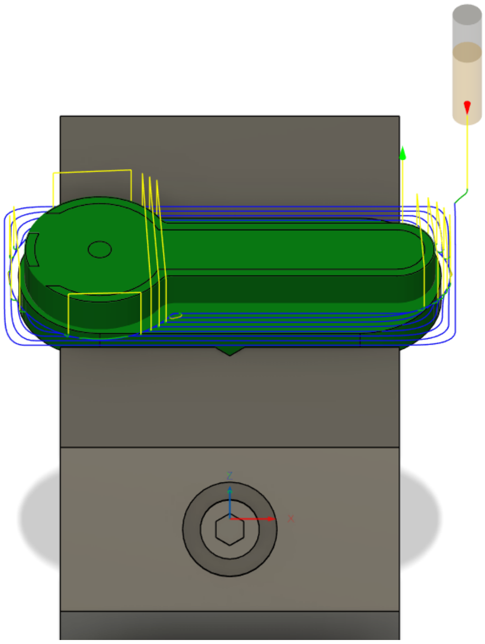


26. Under the *Heights* tab set the bottom height to the floor of the part shown in blue.

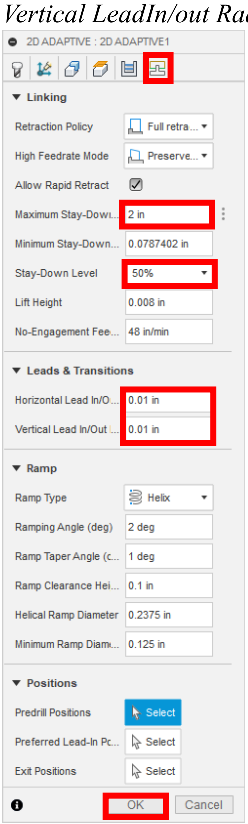


27. Under the *Passes* tab, set the *Optimal Load* to 0.055 in. Turn *Stock to Leave* on and set *Radial Stock to Leave* and *Axial Stock* to 0.005.

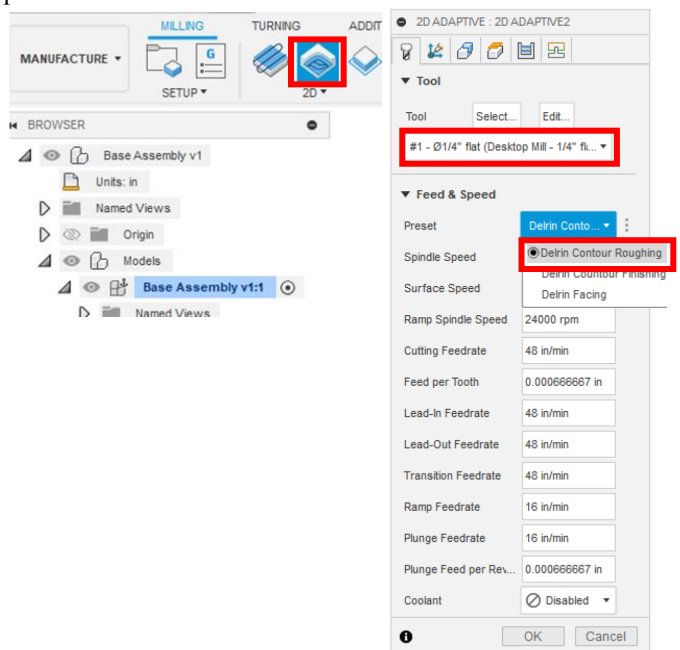




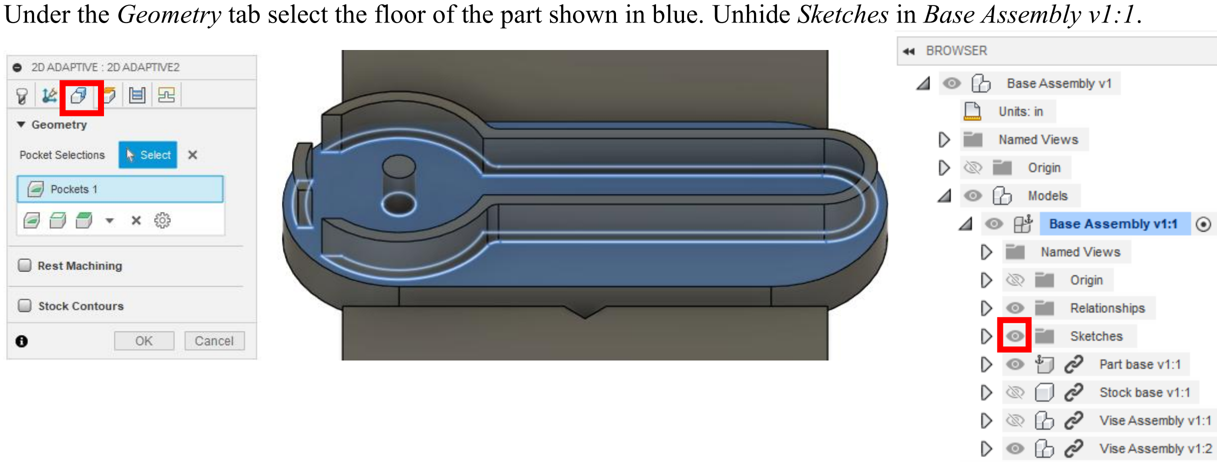
28. Under the *Linking* tab, set *Maximum Stay-Down Distance* to 0.2 in, *Stay Down Level* to 50%, *Horizontal* and *Vertical LeadIn/out Radius* to 0.01 in. Press *OK.*



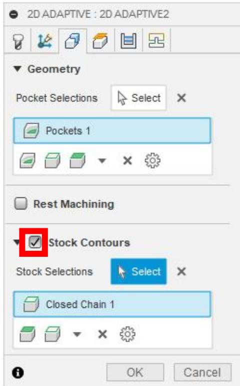
29. Create a *2D Adaptive Clearing* operation. Ensure the same tool is selected. Select the *Delrin Contour Roughing* preset.

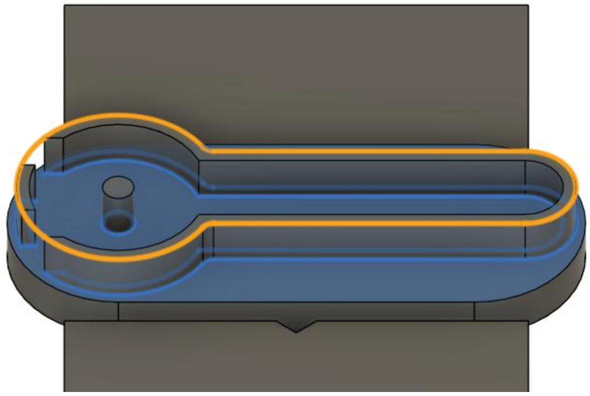


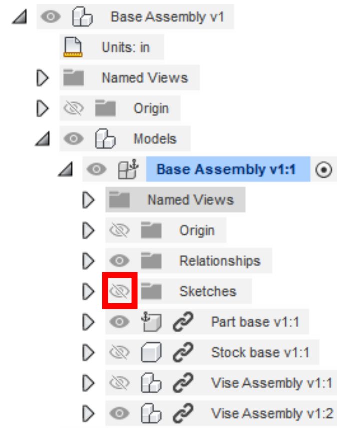
30. Under the *Geometry* tab select the floor of the part shown in blue. Unhide *Sketches* in *Base Assembly v1:1*.



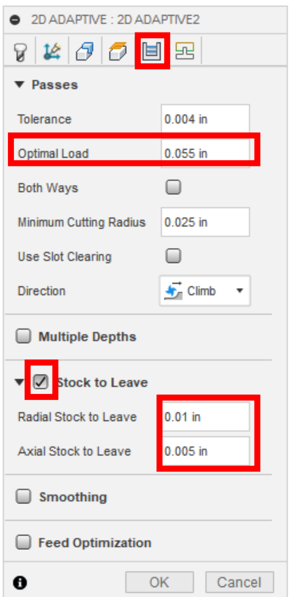
31. Select the *Stock Contours* box and select the black line on the sketch then hide the sketch again. The stock contour should be shown in yellow.

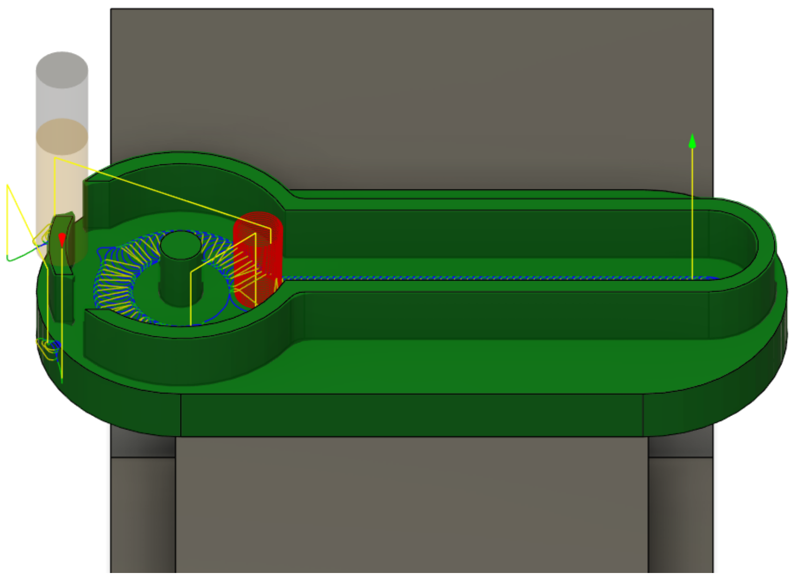




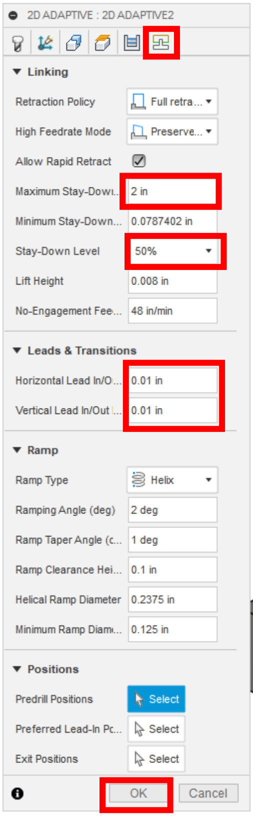


32. Under the *Passes* tab, set the *Optimal Load* to 0.055 in. Turn *Stock to Leave* on and set *Radial Stock to Leave* to 0.01 in and *Axial Stock to Leave* to 0.005 in. Press *OK.*

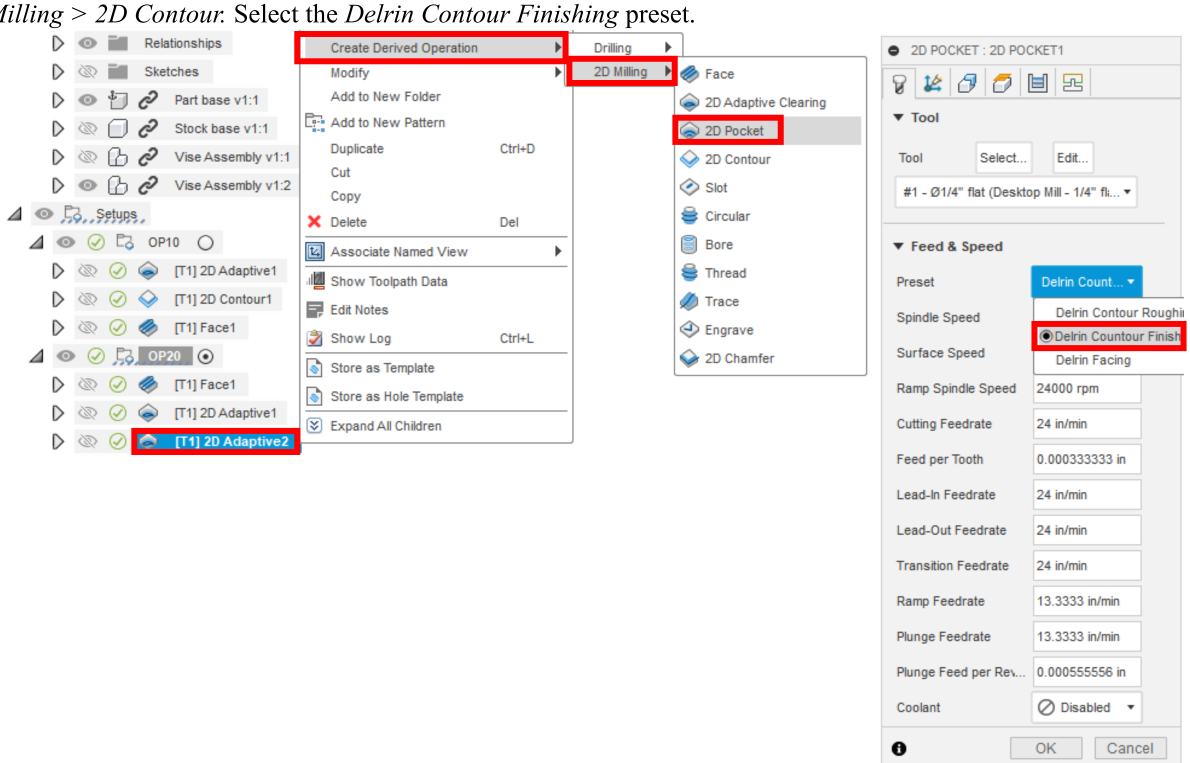




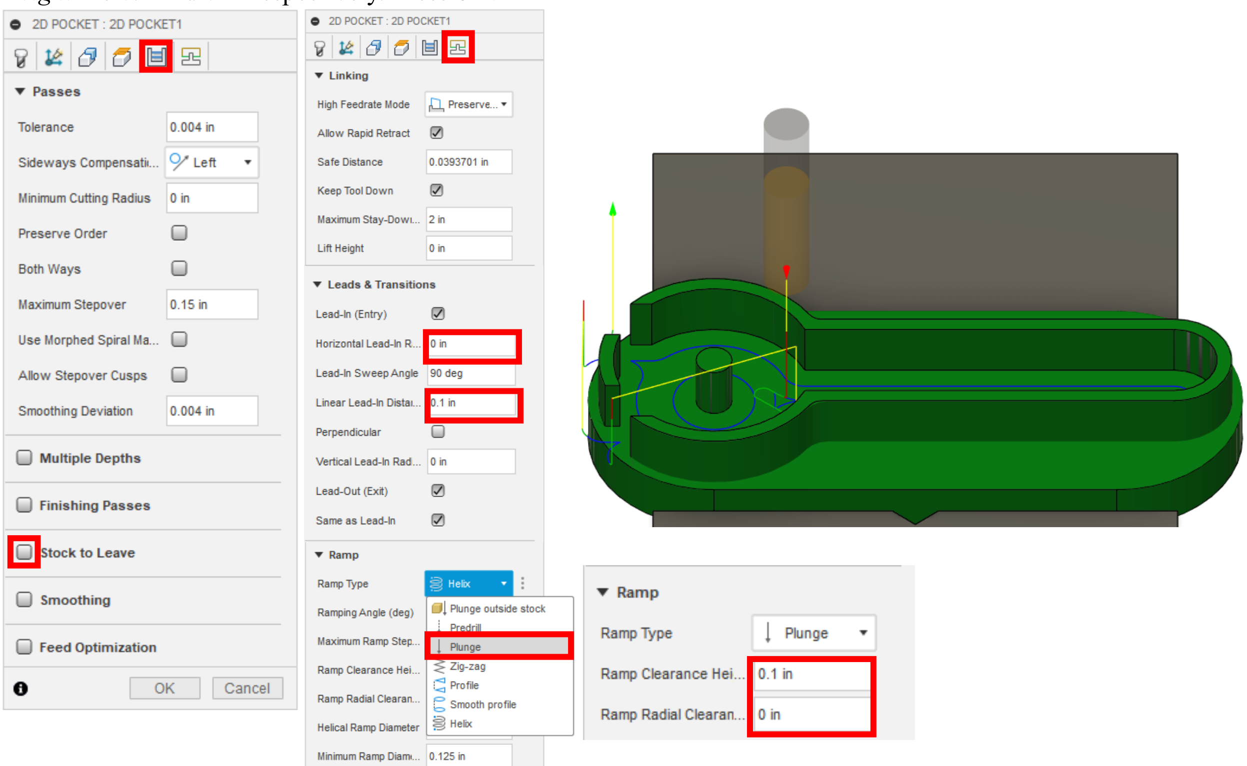
33. Under the *Linking* tab, set *Maximum Stay-Down Distance* to 0.2 in, *Stay Down Level* to 50%, *Horizontal* and *Vertical LeadIn/out Radius* to 0.01 in. Press *OK.*



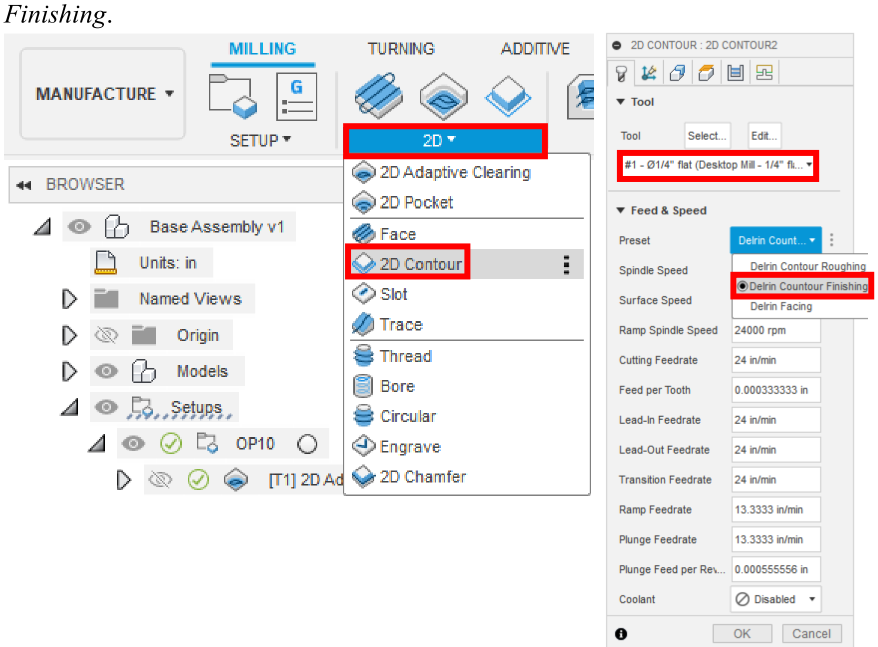
34. Right click the *2D Adaptive Clearing* tool path that was just created. Select *Create Derived Operation > 2D Milling > 2D Contour.* Select the *Delrin Contour Finishing* preset.



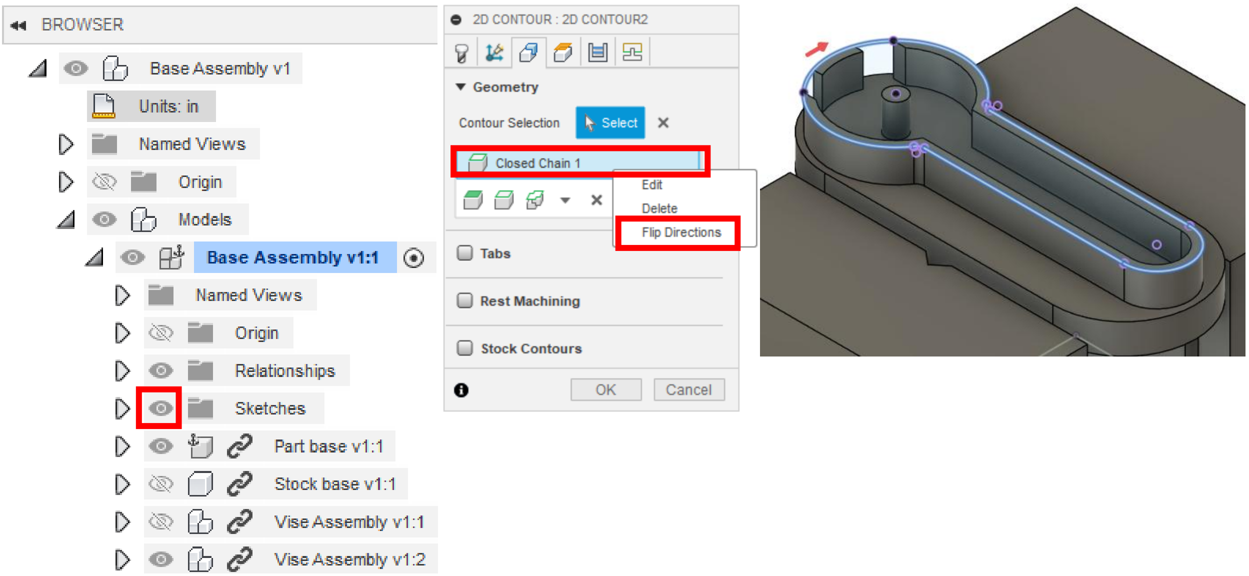
35. Under the *Passes* tab, turn off *Stock to Leave*. Under the *Linking* tab, set *Horizontal Lead-In Radius* to zero, *Linear Lead-In Distance* to 0.1 in, and change the *Ramp Type* to Plunge. Ensure *Ramp* and *Radial Clearance Height* are 0.1 and 0 in respectively. Press *OK*.



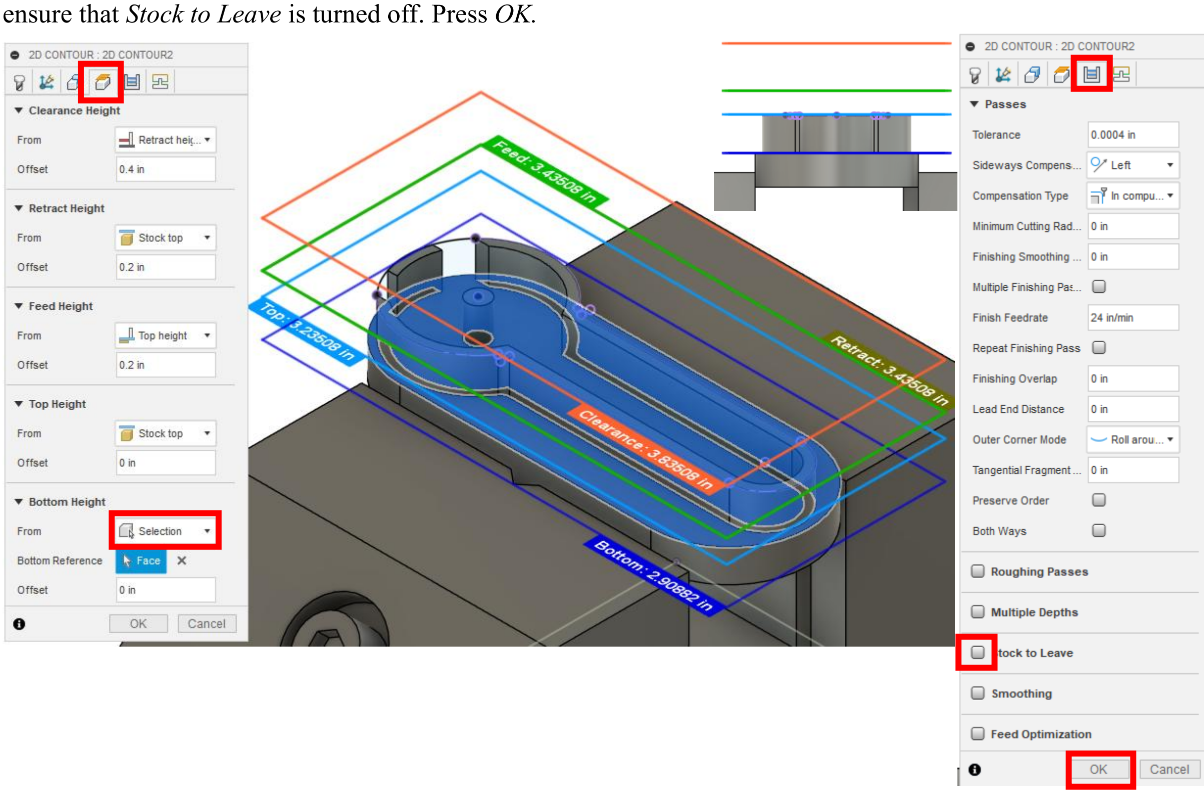
36. Create a *2D Contour* toolpath. Make sure the *1/4" flat endmill* is selected and set *Preset* to *Delrin Contour Finishing*.



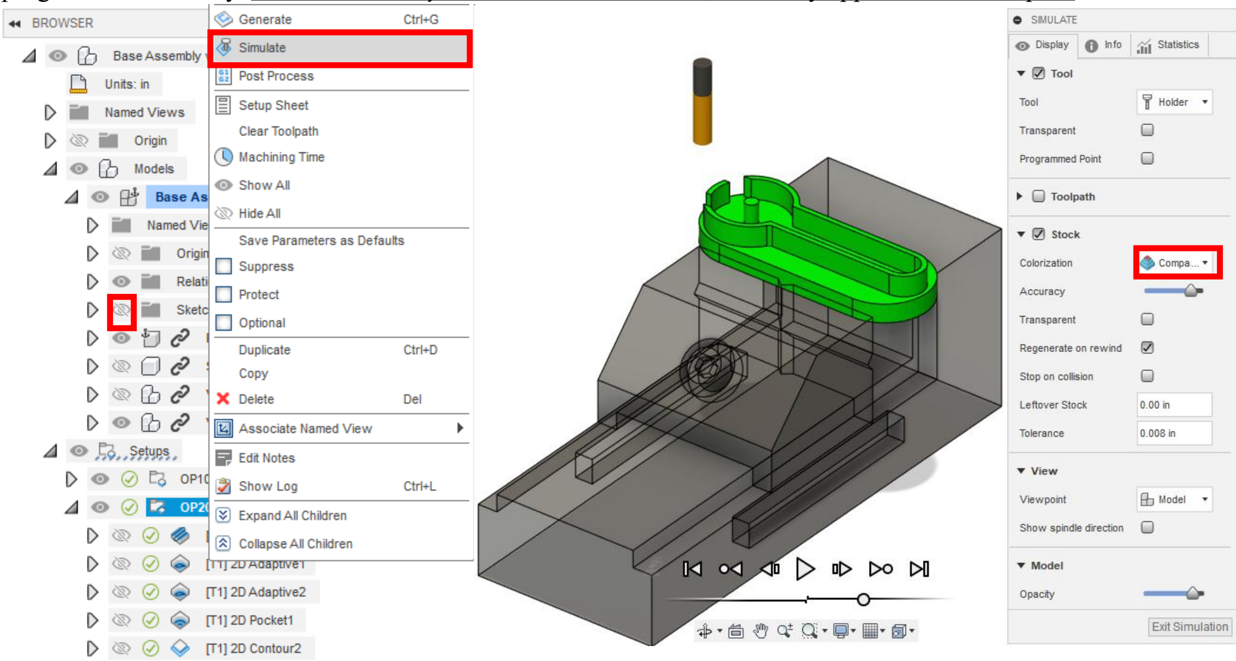
37. In browser, unhide *Sketches* under *Base Assembly v1:1*. Under *Geometry*, select the sketch outlining the outer geometry of the Base as shown. If the red arrow is on the inside of the geometry, right-click on the selection and click *Flip directions*.



38. In the *Heights* tab, change the *Bottom Height* to *Selection*. Select the face shown below. In the *Passes* tab, ensure that *Stock to Leave* is turned off. Press *OK*.



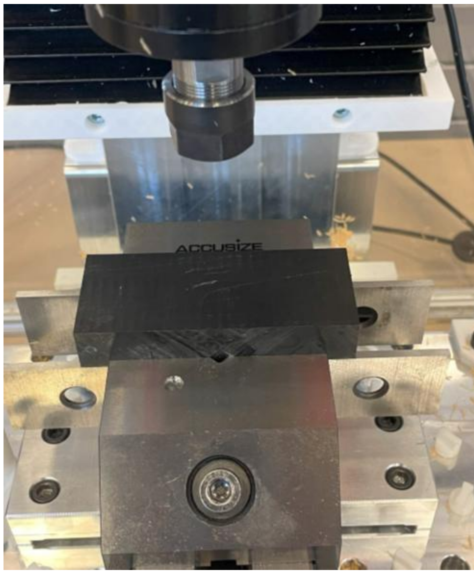
39. *OP20* should now be finished. Hide *Sketches* in *Base Assembly v1:1*. Right click on *OP20* and select *Simulate*. Under *Colorization* choose *Comparison.* Press the play button at the bottom center of the window. Adjust the *Accuracy* slider until simulation runs smoothly. Once the simulation is complete, the part should be green if programmed correctly. <u>Note: If *Accuracy* is set to the lowest value, blue may appear around the part.</u>

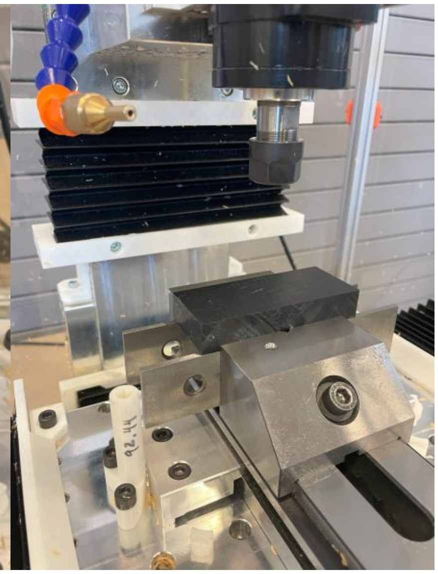


IV: Setup and Probing

OP10

1. Place the base stock centered on the vice jaws on 1 1/8” parallels. The stock should be resting with the widest face down on the parallels and the longest edge along the X-axis as seen below.



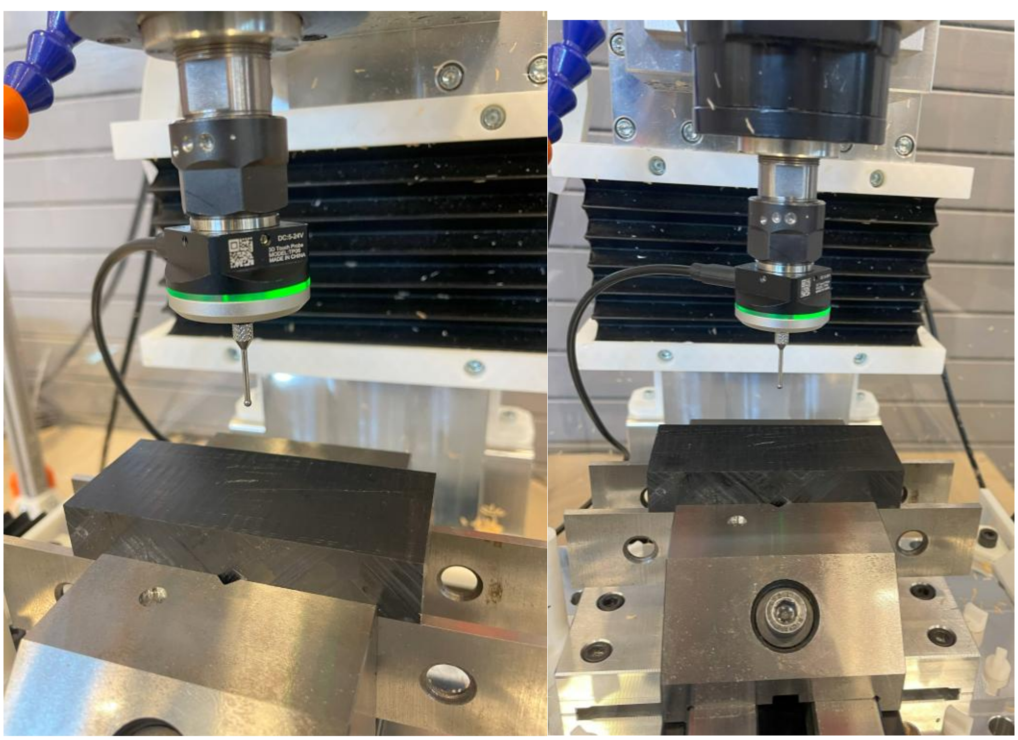


2. Use a 6mm Allen key to tighten the bolt in the center of the front vise jaw (if it is a different vise style, follow manufacturer instructions to clamp the part). This should close on the part and clamp tightly. As you tighten the vise, apply force down on the part. This will make sure that the part is evenly seated on the parallels. Check this by gently shaking the parallel; if it shakes noticeably against the part, try again (NOTE: an uneven surface may not seat perfectly against a flat parallel).

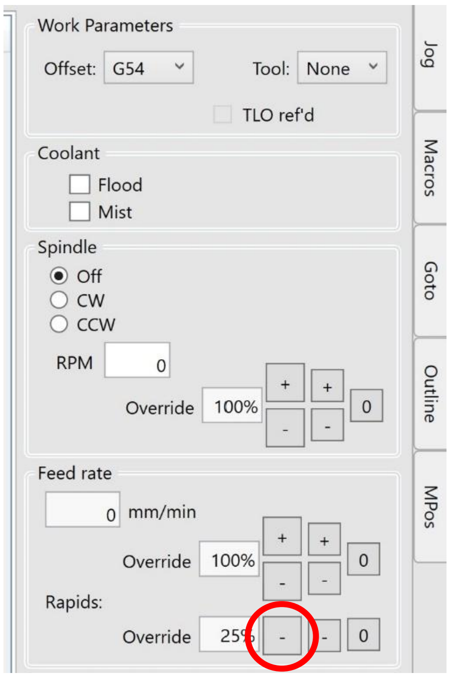
3. Gather the probe and attach the probing tip. Slide the probe into the spindle with a ¼” collet and tighten by hand. This should hold the probe safely. Be sure that the spindle is powered off by the emergency stop or door switch.


4. With the electrical box on and connected to both the computer and machine tool, plug the probe into the electrical box and GENTLY displace the probe tip. The light should change from green to red and the computer software should show a “P” alarm while the probe tip senses contact. Both conditions must be met to ensure probe operation.

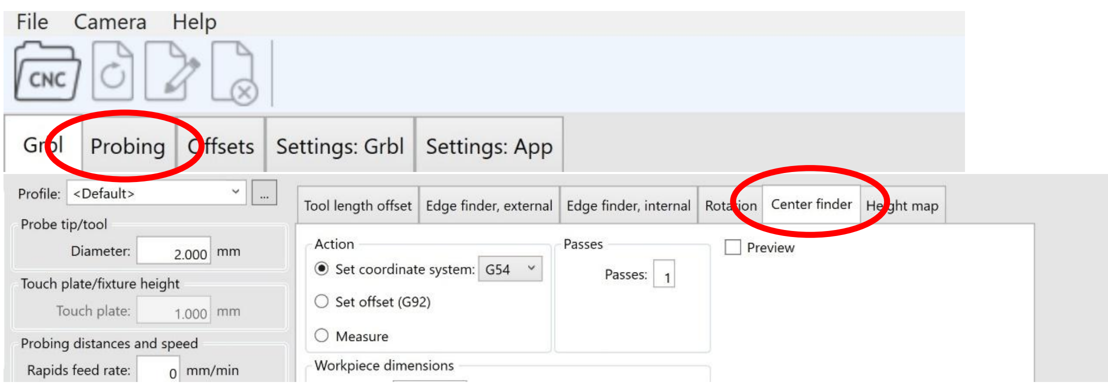
5. Carefully jog the probe over the part. Do not allow the probe tip to touch any surface while jogging, as excessive force or deflection can permanently damage the probe tip. Carefully jog the probe until it is as centered as visually possible over the part and 5-7 mm from the top surface of the part.



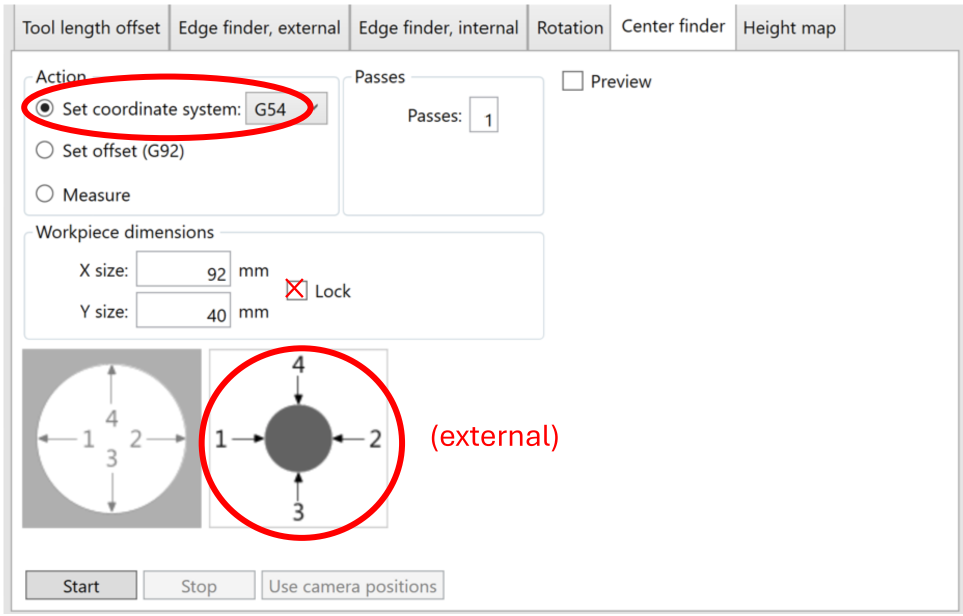
6. Before probing, reduce the rapids rate to 25%. Rapid movement rate can be changed with the minus sign buttons in the “Rapids” section shown below. This is an important step. This is done so that any errors in probing can be safely caught before any damage is done.
NOTE: If using a different version of ioSender, there may be a “slider” instead of buttons. Slide bar to approximately 25% rapid rate.



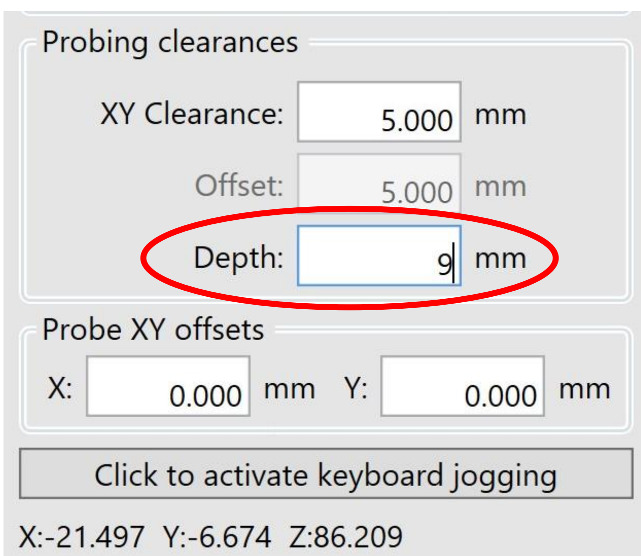
7. Move to the “Probing” tab in the ioSender software. From there, move to the “Center finder” option.



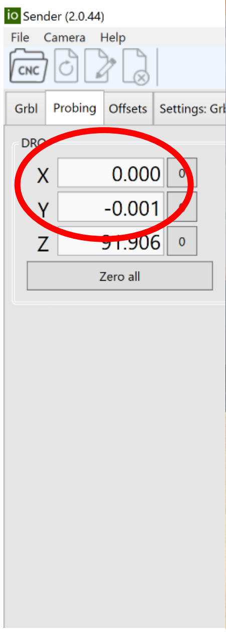
8. Ensure that the “Action” is set to “Set coordinate system” in G54. Uncheck the “Lock” box. Measure the stock and type in the workpiece dimensions in millimeters. This should be roughly 92mm in the X-direction and 40mm in the Y-direction, but stock sizes may differ. Select the external option. Select the external bore option for probing the outside surfaces of the stock.

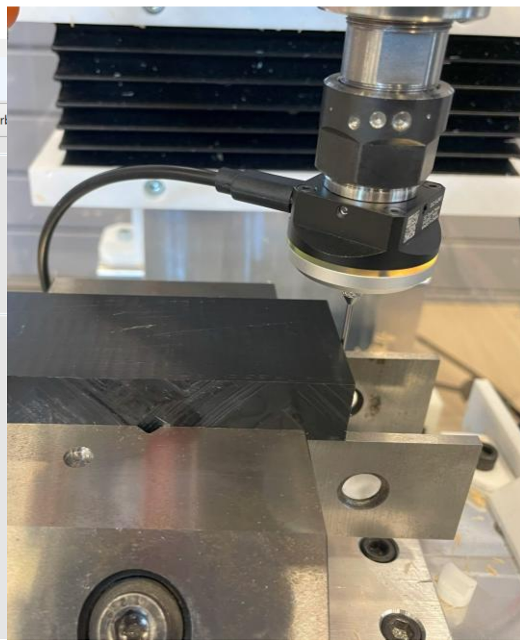


9. In the “Probing clearances” section, set the probing depth. This value will determine how far down the probe travels to touch the workpiece. It is equal to the distance between the current probe tip position and the place on the part that will be probed-- it may be necessary to roughly measure this distance before probing to ensure correct positioning and mitigate errors.



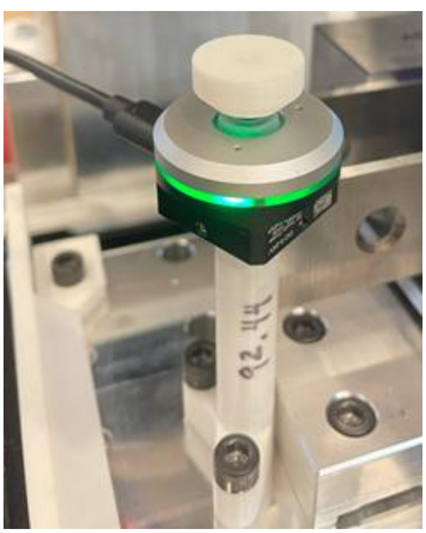
10. Double check inputs. Errors are much more difficult to catch quickly in operation. Press “Start” and watch the probing cycle very closely. Emergency-stop the machine if anything unexpected occurs and safely try again. The machine should move across the part to touch each side once. This will set the X and Y offsets for a coordinate system centered on the rectangular block. The X- and Y- offsets should be zeroed (this is sometimes displayed on ioSender with ±0.001”).





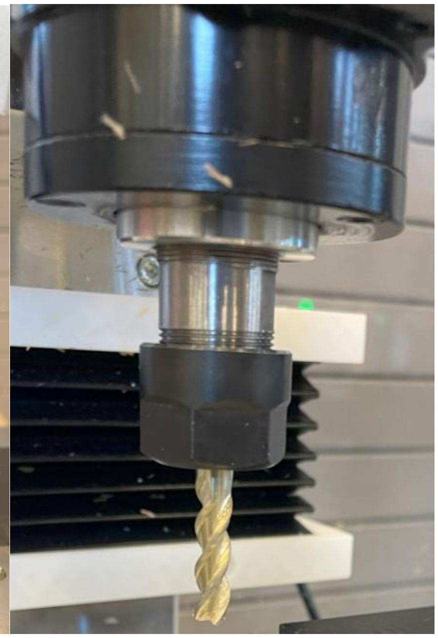

11. We will now set the Z height for the coordinate system. This is based on the surface of the X-carriage and will include the tool offset. Jog the probe safely away from the part and remove it from the spindle.

12. Gently replace the probe tip with the probe touch plate. Insert the shaft of the probe into the holder on the left side of the vise. If the signal cable was disconnected, plug it into the probe and test again to be sure the signal is working.

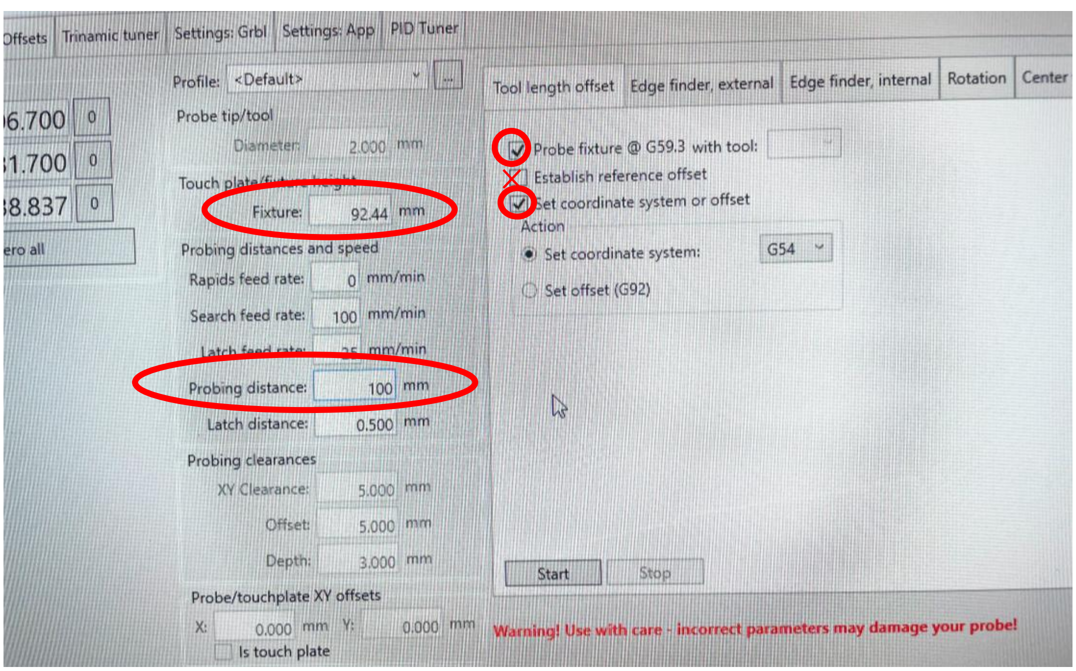


13. **Never handle tools in the enclosure while the spindle is powered.**

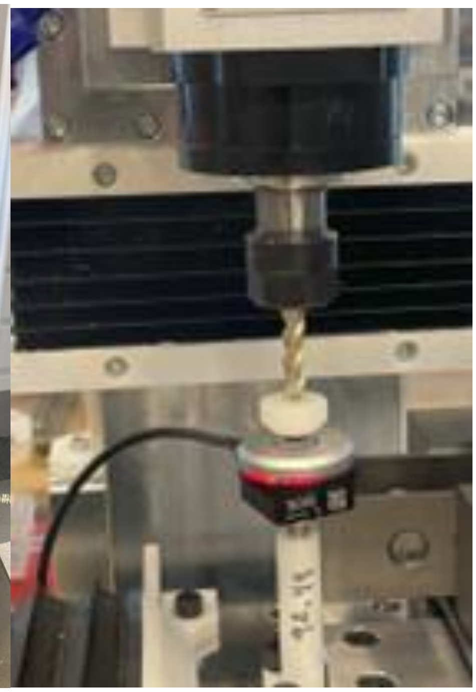
Insert the ¼” endmill into the spindle and secure tightly using the provided wrenches. The tool should protrude just past the start of the tool shank.



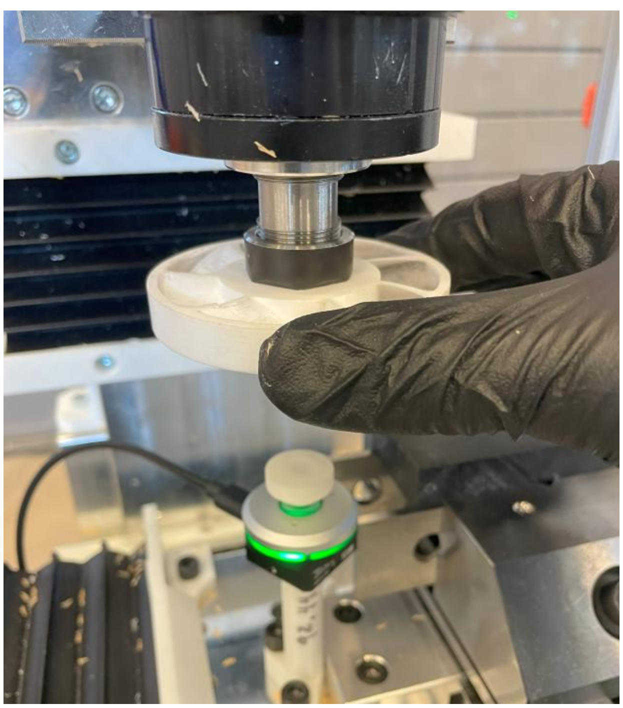
14. Jog the tool so that there are no obstructions between the tool and touch probe. Move back to the “Probing” tab and select “Tool length offset”. Check the “Probe fixture @ G59.3 with tool” box, uncheck the “Establish reference offset” box, check the “Set coordinate system or offset” box, and set the touch plate height to 92.44 mm (or the otherwise determined accurate distance from the touch plate to the X-carriage surface). Then set the probing distance to 100 mm.



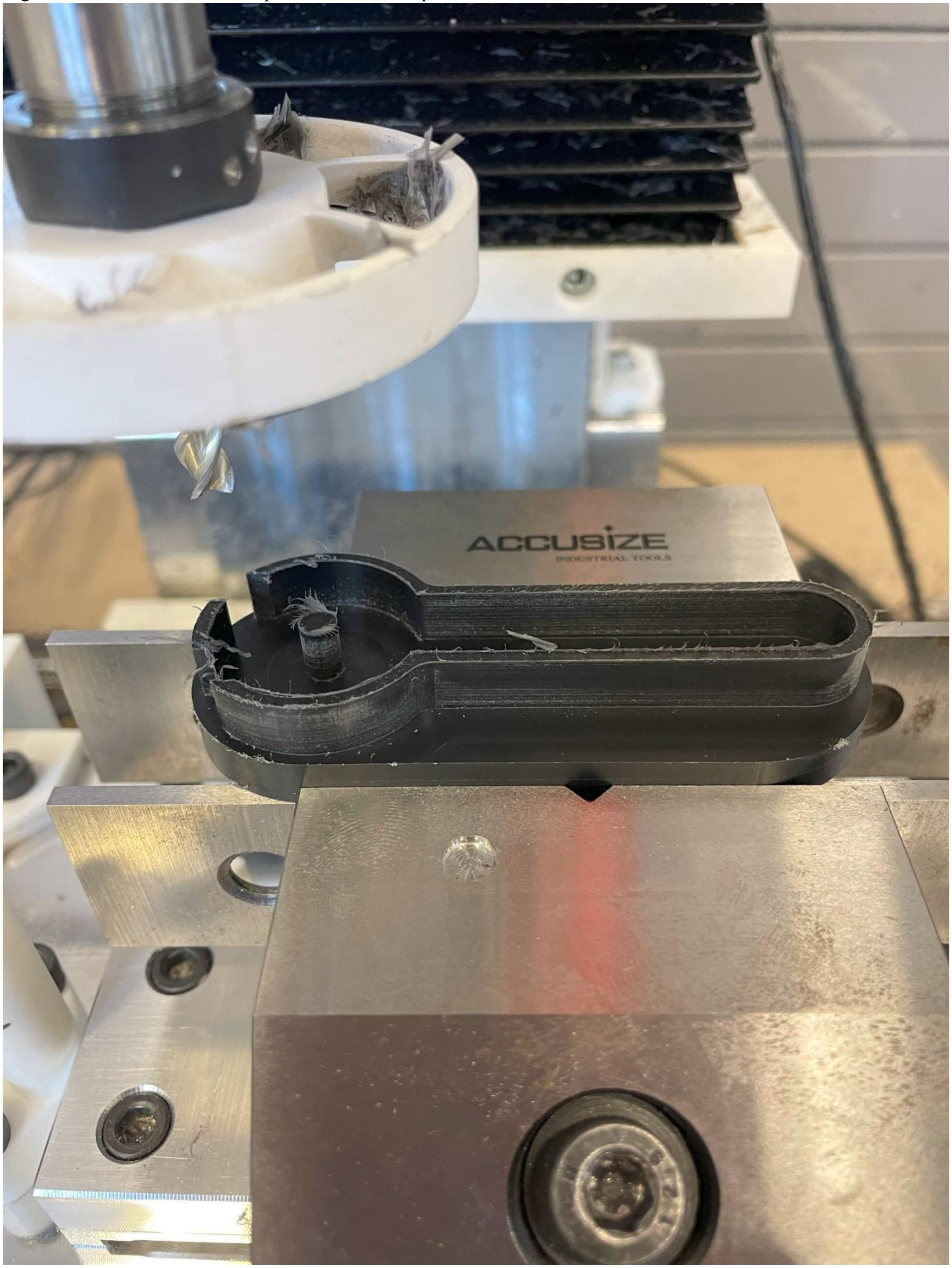
15. Double check all parameters and press start. The tool should slowly approach the touch plate and gently press it twice. This will set the Z-coordinate at the bottom of the vise / top of the X-carriage.



16. Remove the probe completely from the enclosure. If the machine tool is not equipped with compressed air, slide the plastic chip fan onto the spindle over the tool as seen below.



17. Jog the tool so that it starts the operation **above** the part.



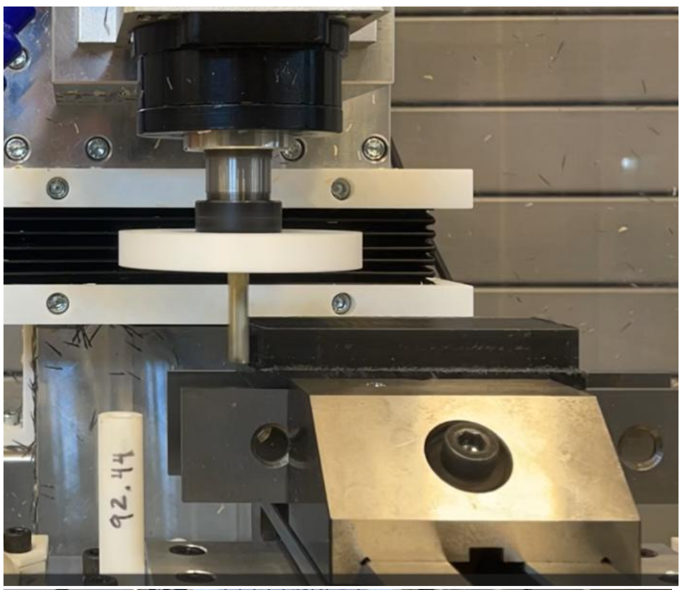
The machine tool should now be ready to run the first operation.

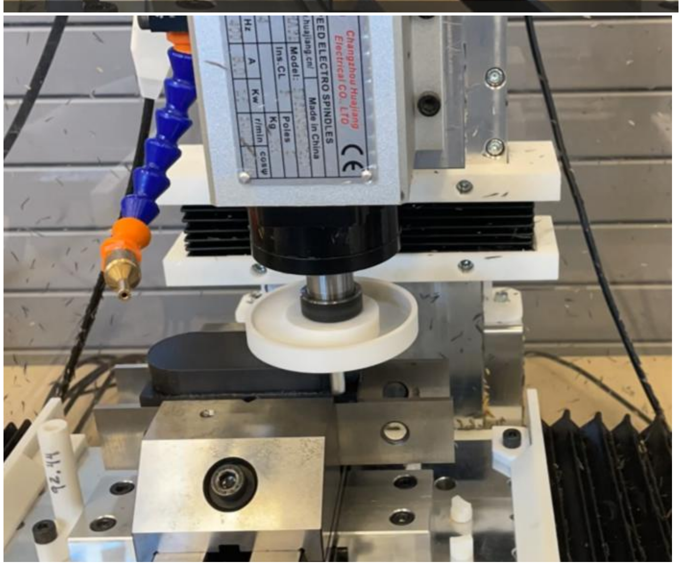
REVIEW AND FOLLOW GENERAL INSTRUCTIONS AND CHECKLIST.

Be ready to feed hold the machine IMMEDIATELY after pressing cycle start. This will allow the spindle to reach the desired speed before milling begins (this usually takes no longer than 5 seconds – the spindle should be making a constant sound and not “ramping up”).
ALWAYS ALLOW THE SPINDLE TO GET UP TO SPEED BEFORE ENTERING A CUT.

After pressing feed hold and letting the spindle reach its target RPM, you can resume the operation with cycle start.

18. Watch the entire operation carefully.

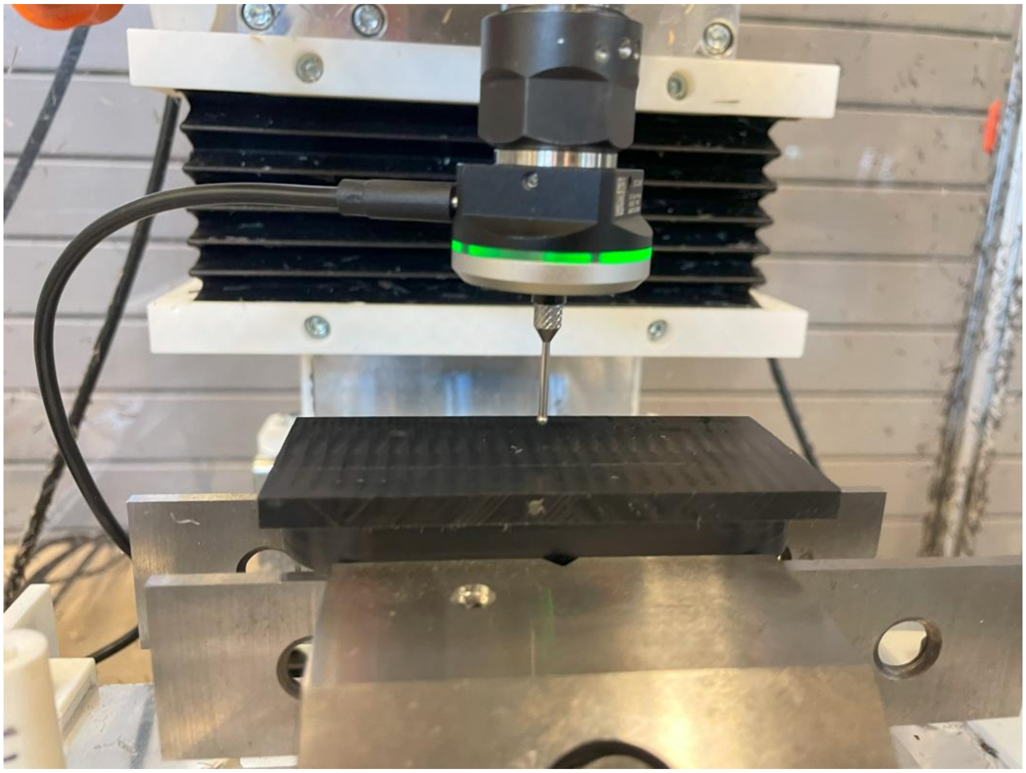




OP20

19. Once OP10 is complete, jog the tool out of the way. Stop the machine and open the door. Clear any chips as necessary. Carefully slide the chip fan off the spindle. Remove the ¼" endmill with the wrenches.

20. Use the Allen key to loosen the vise jaw and remove the cut part. It may be necessary to clean up edges with a file or deburring tool.

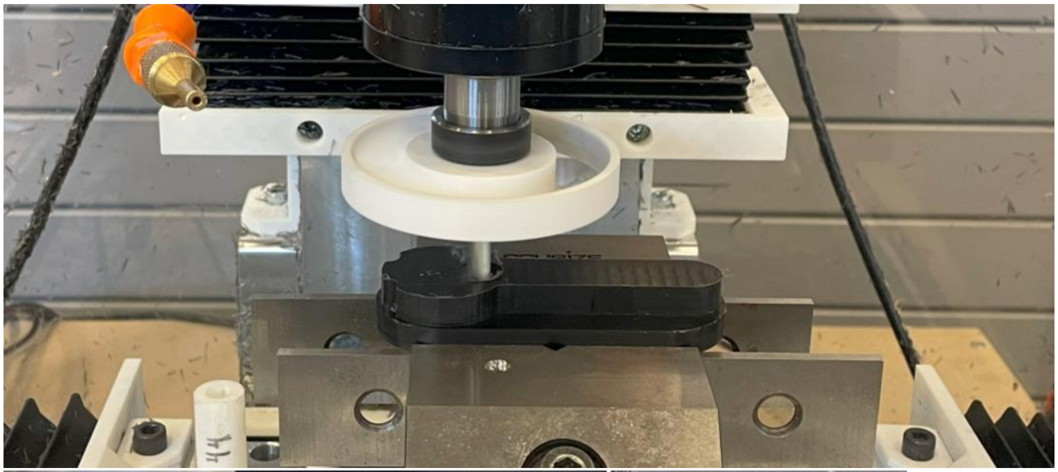


21. Replace the part upside down, on the same parallels, and clamp down on the newly cut surface in the same orientation as before.

22. Follow steps 5-15 to probe the part and ¼" tool. The X- and Y- measurements for stock probing remain the same. For this setup, ensure that the probing depth is set so that the flat faces of the stock contact the probe tip and not the probe shaft while probing.

23. Repeat steps 16 and 17.

24. Watch the entire operation carefully.



25. Once OP20 is complete, jog the tool out of the way. Emergency stop the machine and open the door. Clear any chips as necessary. Carefully slide the chip fan off the spindle. Remove the ¼" endmill with the wrenches.

26. Use the Allen key to loosen the vise jaw and remove the cut part.